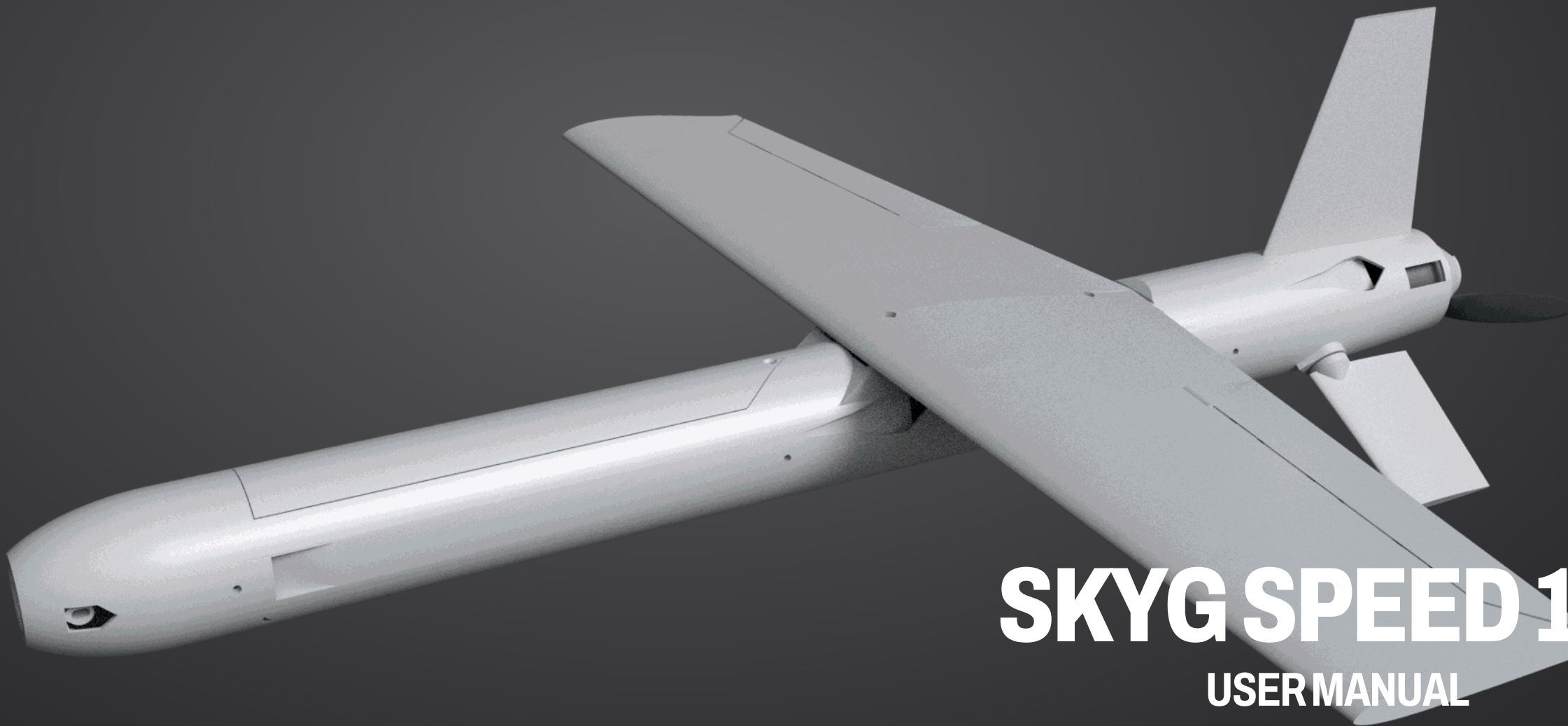


**SKYG**



# **SKYG SPEED 120**

**USER MANUAL**

V.1 · C7-2 · 2026-10-02

# PACKAGE CONTENTS

## Manual

SkyG Speed 120 user manual (this document).

## STEP

Every printed part and the full assembly, for editing in any CAD program.

## ArduPilot

Setup document and a starting .param file for the Matek H743-WLITE, and the rpm-governor script.

## Engineering

Design record, parts list (BOM.csv), test plan and verification report.

## STL

Every printed part, in millimetres, ready for the slicer (print orientation: pages in this manual).

## Wiring

The wiring diagram as a separate large page.

## Viewer

Interactive 3D model (open the HTML file in a browser): every part exploded and in place, with notes.

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## Assembly guide

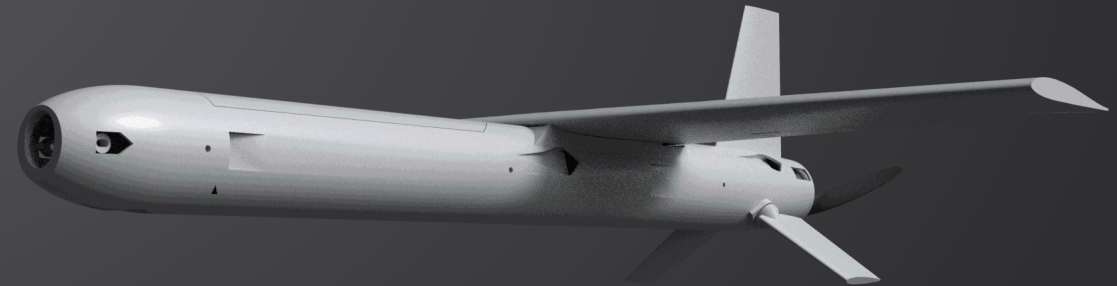
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# SKYG SPEED 120

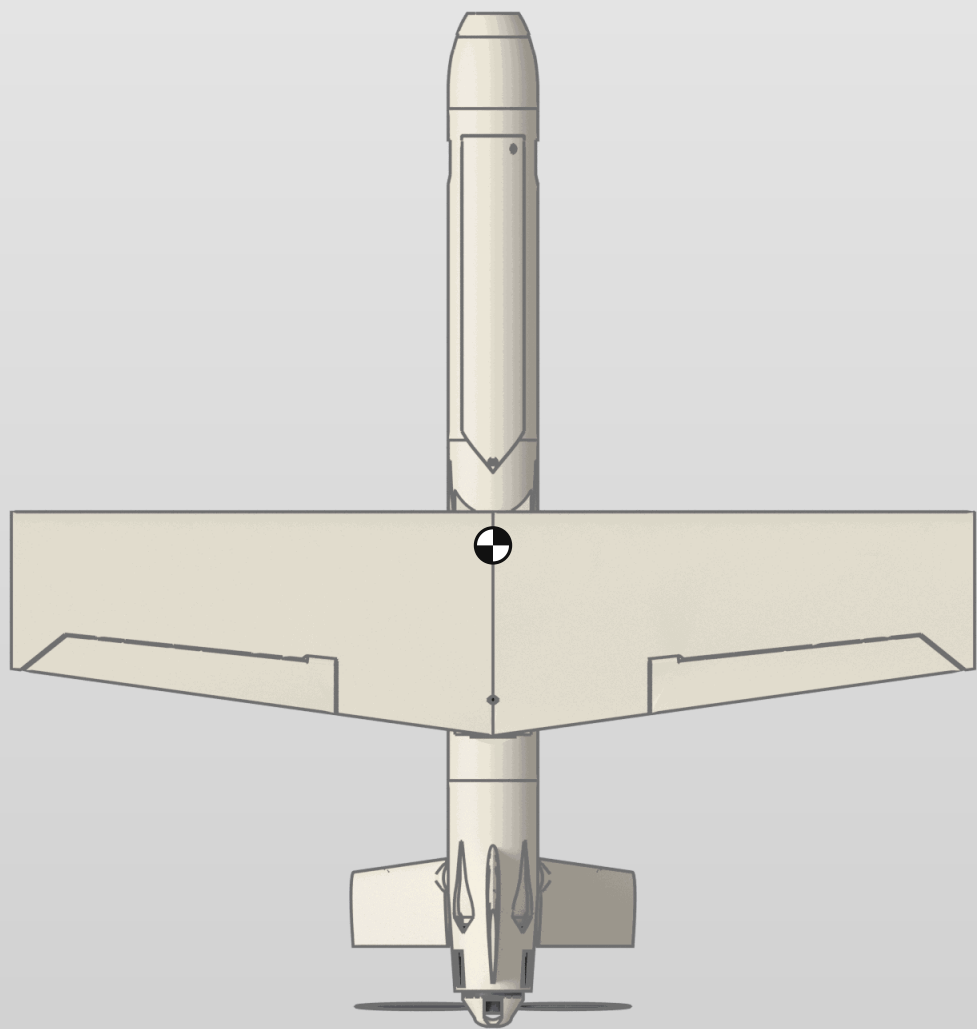
SkyG Speed 120 is a compact, hand-launched speed aircraft: a 3D-printed fuselage in three plug-together modules and a separate camera nose, a 430 mm straight wing on a carbon spar, and an all-moving canted tail driven by two micro servos. A single pusher motor sits in the tail cone behind the tail.

The airframe is printed in foaming ASA Aero on a 0.4 mm nozzle; steel and carbon parts carry the concentrated loads (spar, aileron hinge pins and torque rods, tail shafts). Six interchangeable camera noses take a micro FPV camera, a Raspberry Pi camera or a DJI O4 Air Unit, each with an open or a clear-dome face.

It is built for straight-line speed: about 118 m/s as printed and 120 m/s finished in the model with the clear-dome nose, a 6S pack and an rpm-governed carbon prop, flown up through never-exceed stages to 120 m/s.



# GENERAL AIRCRAFT DATA



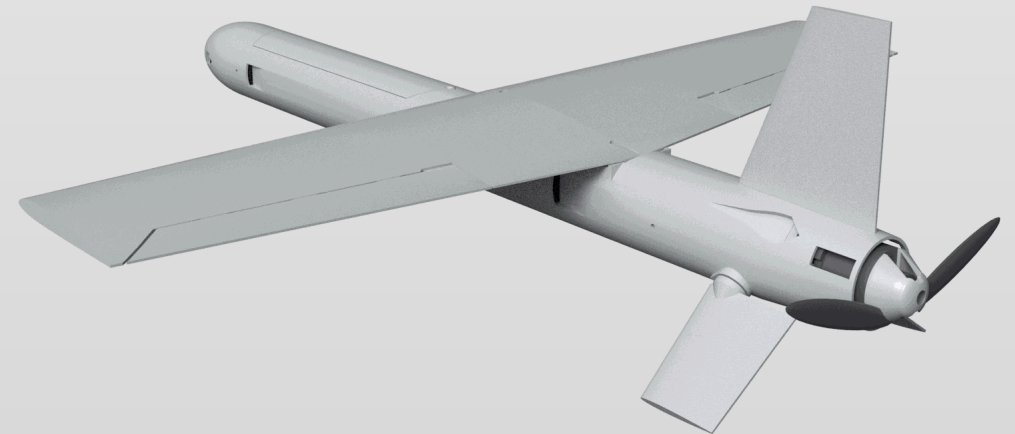
|                   |                                                |
|-------------------|------------------------------------------------|
| Wingspan          | 430 mm                                         |
| Wing area         | 3.65 dm <sup>2</sup>                           |
| Length            | 449 mm                                         |
| Centre of gravity | 15 mm behind the wing leading edge at the root |
| All-up weight     | 400 g (clear-dome nose)                        |
| Stall speed       | 13.2 m/s                                       |
| Top speed (model) | 118–120 m/s                                    |
| Airfoil           | MH45, ailerons 3° reflex                       |
| Root / tip chord  | 100 / 70 mm                                    |
| MAC               | 86 mm                                          |
| Aspect ratio      | 5.1                                            |
| Wing loading      | 107 g/dm <sup>2</sup>                          |

# AERODYNAMIC DESIGN

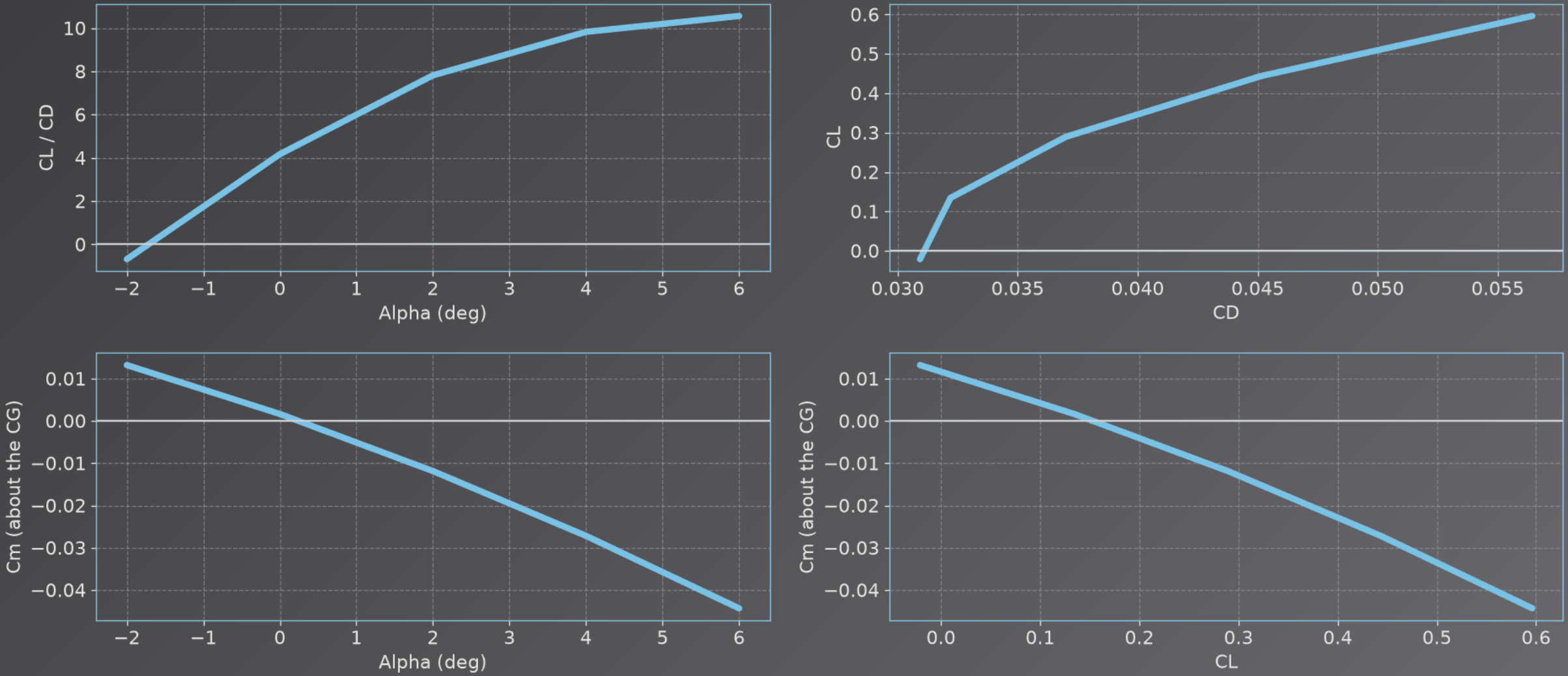
The wing uses the MH45 section, a low-moment reflexed profile, with both ailerons rigged 3° trailing edge up. The design CG is at 10 % of the mean chord, 9.1 % static margin ahead of the neutral point (x 246 mm from the nose datum): stable without a tail download penalty at speed.

Pitch and yaw come from two all-moving tail panels at 35° anhedral, mixed like a V-tail, with a fixed fin on top. The Dutch roll is lightly damped at low speed, so the flight controller's yaw damper is part of the setup.

Flutter of the wing, ailerons and tail panels was analysed (strip theory with the CAD-derived stiffness and masses): the lowest predicted instability is 168 m/s, above 1.2 × the 120 m/s top stage. The test plan opens the speed envelope in stages and measures the stiffness the margin depends on.



# AERODYNAMIC CHARACTERISTICS



Whole aircraft, VSPAERO vortex lattice with the fuselage panels; best L/D 10.6 at 6.0°;  $C_m$  about the design CG.

# PERFORMANCE AND SPEED STAGES

|                              |                                                    |
|------------------------------|----------------------------------------------------|
| Motor                        | T-Motor F50 2207 2150KV                            |
| Propeller                    | Gemfan 5x7.5E Vortex                               |
| Battery                      | BetaFPV LAVA II 3S 680 95C LiHV                    |
| Static thrust                | 1458 g at 60 A (thrust / weight 3.66)              |
| rpm governor / current limit | 38,000 rpm max / 60 A                              |
| Top speed                    | 118 m/s as printed, 120 m/s finished               |
| Full-throttle time           | about 0.6 min per pack pair (passes of $\leq 8$ s) |

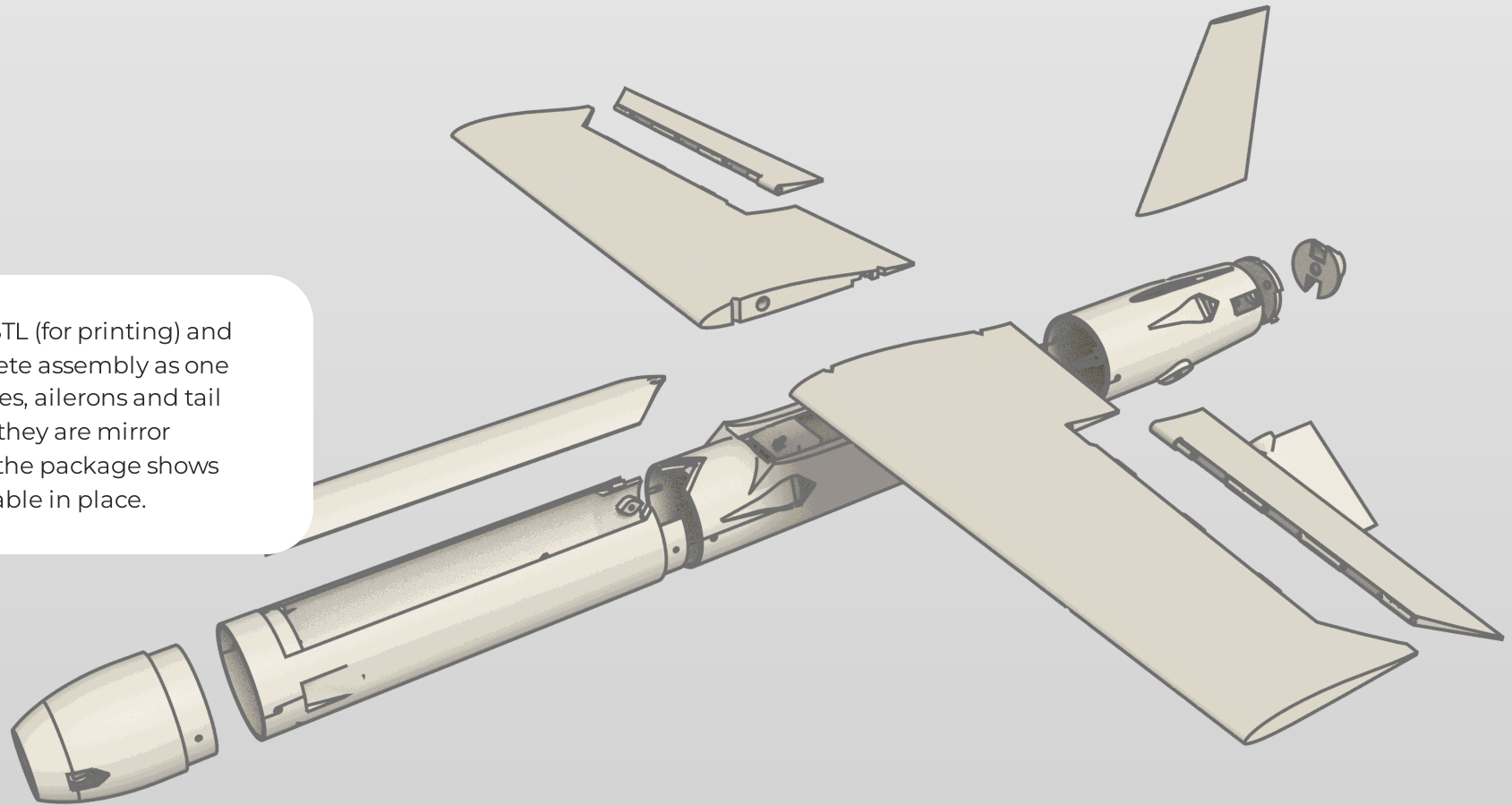
| Stage Vne | GOV_RPM (full stick) | Expected level speed |
|-----------|----------------------|----------------------|
| 30 m/s    | 10,400 rpm           | 30 m/s               |
| 35 m/s    | 12,000 rpm           | 35 m/s               |
| 40 m/s    | 13,500 rpm           | 40 m/s               |
| 45 m/s    | 15,000 rpm           | 44 m/s               |
| 50 m/s    | 16,600 rpm           | 50 m/s               |
| 55 m/s    | 18,100 rpm           | 54 m/s               |
| 60 m/s    | 19,700 rpm           | 60 m/s               |
| 65 m/s    | 21,200 rpm           | 64 m/s               |
| 70 m/s    | 22,700 rpm           | 69 m/s               |
| 75 m/s    | 24,300 rpm           | 74 m/s               |
| 80 m/s    | 25,800 rpm           | 79 m/s               |
| 85 m/s    | 27,300 rpm           | 84 m/s               |
| 90 m/s    | 28,800 rpm           | 89 m/s               |
| 95 m/s    | 30,300 rpm           | 94 m/s               |
| 100 m/s   | 31,900 rpm           | 99 m/s               |
| 105 m/s   | 33,400 rpm           | 104 m/s              |
| 110 m/s   | 34,900 rpm           | 109 m/s              |
| 115 m/s   | 36,400 rpm           | 114 m/s              |
| 120 m/s   | 37,900 rpm           | 118 m/s              |

No airspeed sensor: each stage is a GOV\_RPM value flown at full stick (switch on STAGE). Level runs; GPS two-way passes.

# FILES



Every printed part is supplied as STL (for printing) and STEP (for editing), plus the complete assembly as one STEP file. Left and right wing halves, ailerons and tail panels are separate files because they are mirror images. The interactive viewer in the package shows every part, purchased item and cable in place.



# COMPARTMENT LAYOUT

## BATTERY BAY

6S as two 3S packs side by side, 81 × 31 × 20 mm together; slides x 108–120 for balance; rear stop at x 160

## WING BAY

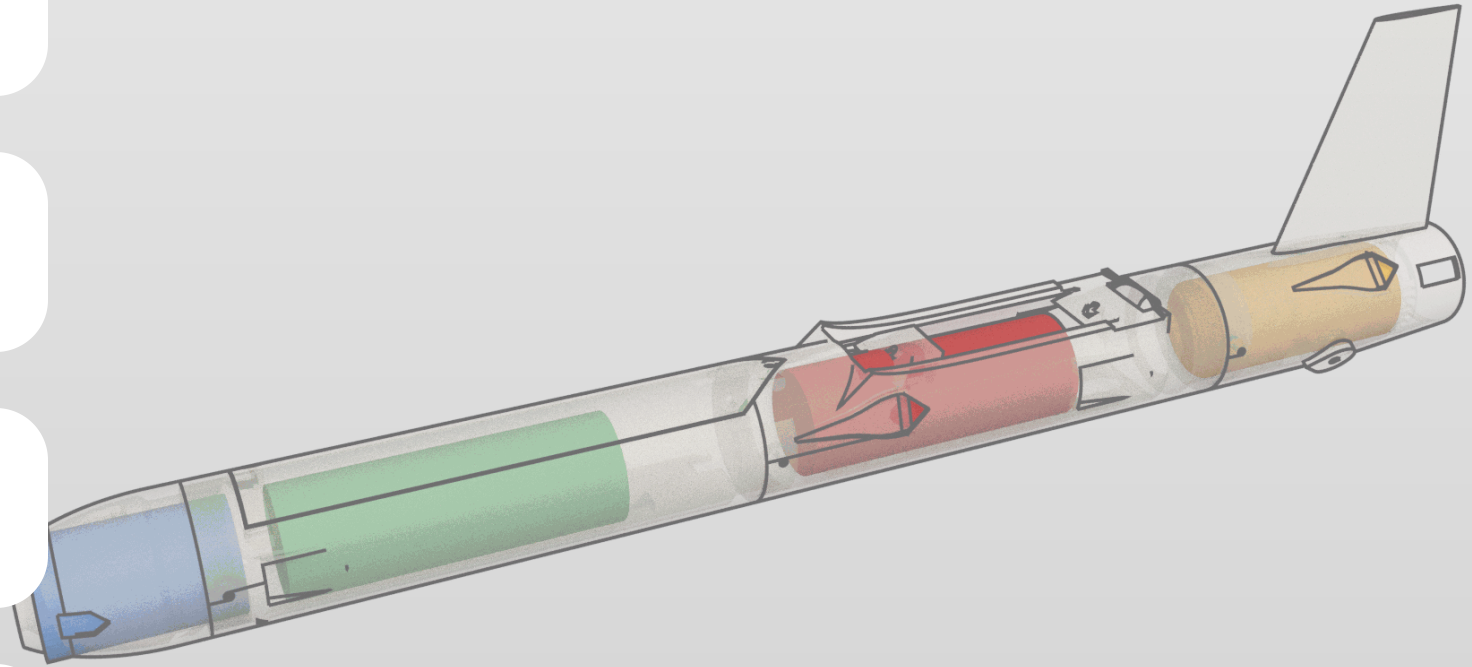
Flight controller stack with the ESC on top, receiver, aileron servo; reached with the wing off

## TAIL MODULE

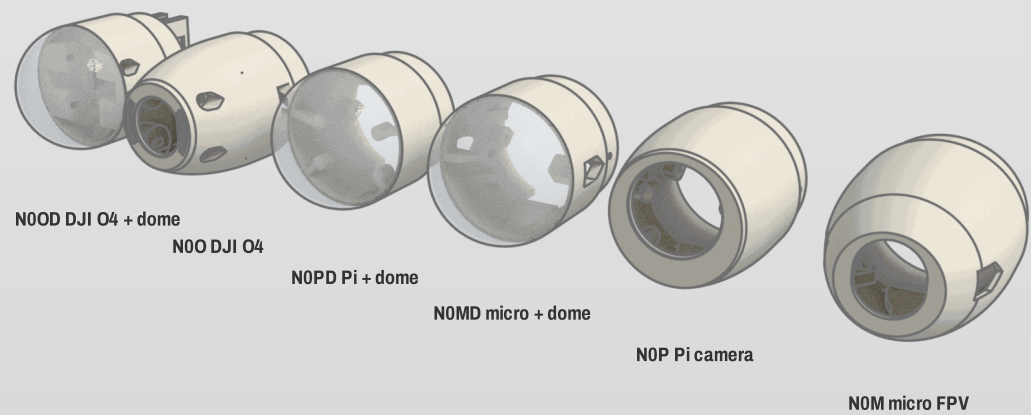
Two tail servos, pushrods, horns and the motor; plugs off at the MR30 and servo plugs

## NOSE BAY

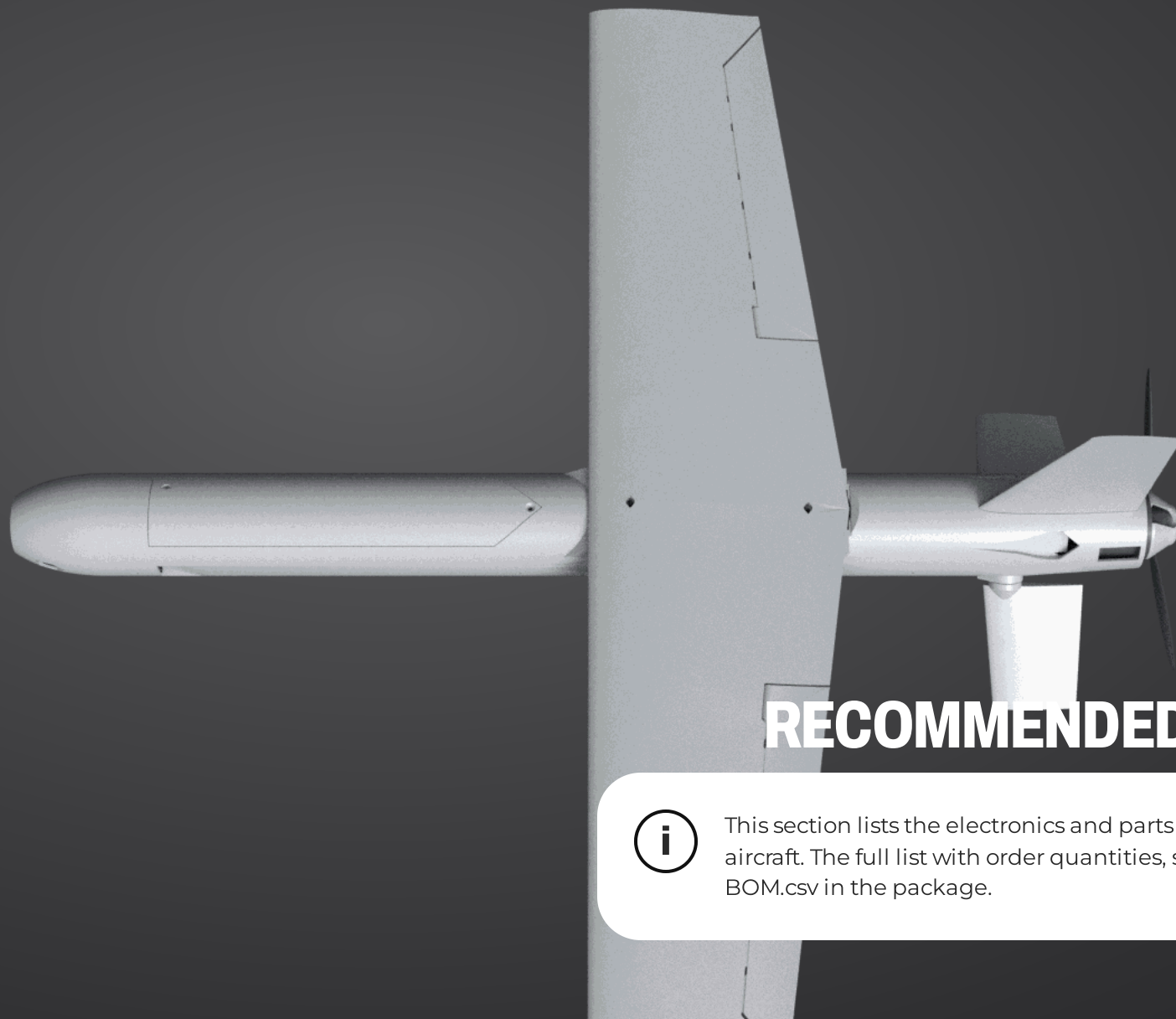
Camera and VTX (or the DJI O4 unit) in the removable nose



# CAMERA NOSE VARIANTS



| Nose | Camera                                                     | Battery centre for balance            |
|------|------------------------------------------------------------|---------------------------------------|
| N0M  | 19 mm micro FPV camera + 20 × 20 VTX                       | x 120 mm                              |
| N0P  | Raspberry Pi Camera Module 3 (test option)                 | x 109 mm                              |
| N0MD | micro camera behind a 40 mm dome, nano VTX under the hatch | x 125 mm (at its stop + tail ballast) |
| N0PD | Pi camera behind a dome                                    | x 113 mm                              |
| N0O  | DJI O4 Air Unit (Lite), ducted                             | x 125 mm (at its stop + tail ballast) |
| N0OD | DJI O4 behind a dome                                       | x 128 mm (at its stop + tail ballast) |



## RECOMMENDED EQUIPMENT



This section lists the electronics and parts needed to complete the aircraft. The full list with order quantities, specifications and sources is BOM.csv in the package.

# RECOMMENDED ELECTRONICS

| COMPONENT                   |  | MODEL / SPECIFICATION                                          |
|-----------------------------|--|----------------------------------------------------------------|
| Motor                       |  | T-Motor F50 2207 2150KV                                        |
| Propeller (fixed)           |  | Gemfan 5x7.5E Vortex                                           |
| ESC                         |  | Sequare SQESC 2670 70A (AM32)                                  |
| Flight controller           |  | Matek H743-WLITE                                               |
| GPS + compass               |  | TBS M10Q GPS GLONASS (QMC5883 compass), 13.5 g                 |
| Servos                      |  | 3 × Emax ES9052MD or ES9052MD HV (Promodelhobby, Turkey)       |
| Flight battery              |  | 2 × BetaFPV LAVA II 3S 680 95C LiHV (2-pack)                   |
| Receiver                    |  | RadioMaster RP1 V2                                             |
| FPV camera                  |  | Micro 19 mm FPV camera, e.g. RunCam Phoenix 2 or Caddx Ratel 2 |
| Video transmitter + antenna |  | 20x20 analog or digital VTX with U.FL antenna                  |
| DJI O4 Air Unit (Lite)      |  | DJI O4 Air Unit                                                |

# RECOMMENDED ACCESSORIES

| ITEM                                     | QTY       |
|------------------------------------------|-----------|
| XT30 connector pairs                     | 2         |
| Carbon tube                              | 1         |
| Music wire 3/32 in (2.38 mm)             | 2         |
| Tail pushrods: wire + micro ball links   | 2         |
| Wheel collar for 2.5 mm shaft + horn arm | 2         |
| Nylon screw M3 x 16                      | 1         |
| M3 hex nut                               | 2         |
| M2 x 6 screws                            | 6         |
| M2 x 6 countersunk screws (front hatch)  | 2         |
| Kapton / thin clear tape                 | as needed |
| Adhesives                                | as needed |
| Velcro strap + hook/loop                 | 1         |
| Heat shrink + silicone wire              | as needed |
| Silver steel rod 2.5 mm                  | 2         |
| Brass tube 3 mm                          | 2         |

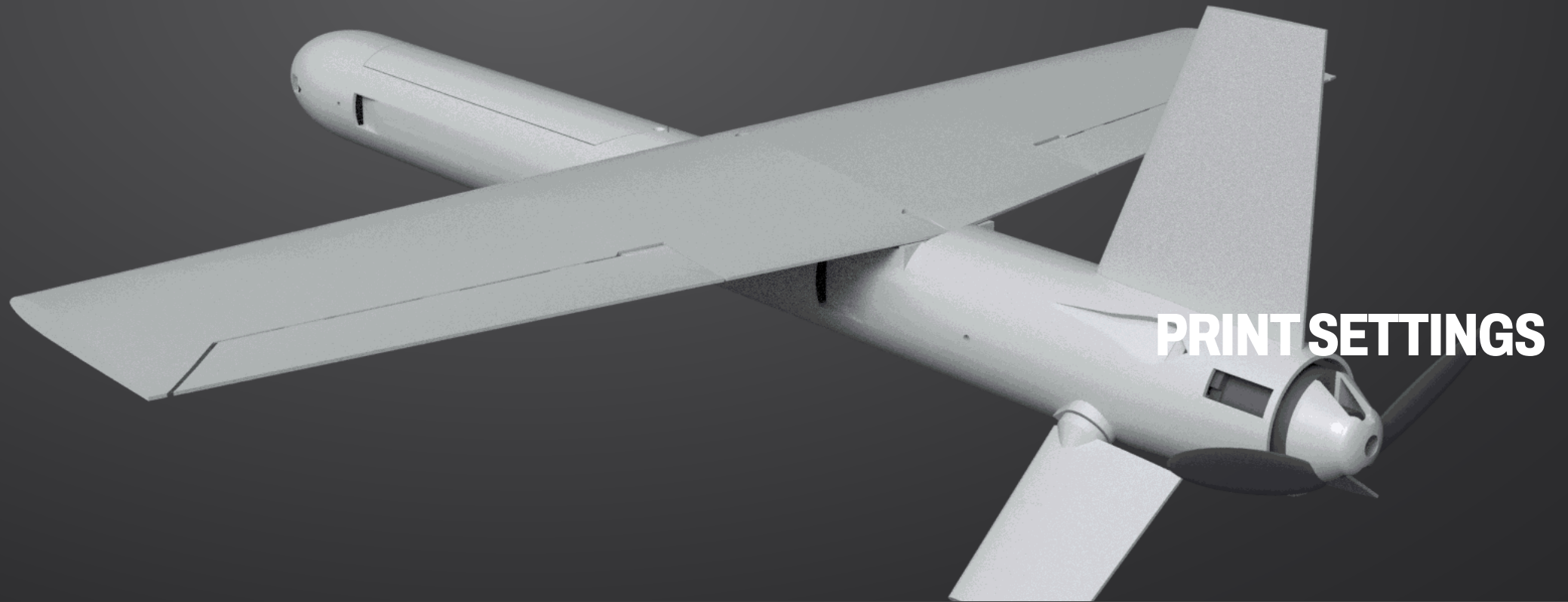
| ITEM                              | QTY  |
|-----------------------------------|------|
| MR30 connector pairs              | 1    |
| Servo extension leads 100 mm      | 2    |
| Motor screws M3                   | 4    |
| Glass skin + finish               | 25 g |
| Nylon screw M3 x 12               | 1    |
| M2 x 5 screws                     | 4    |
| Servo strap / zip tie + foam tape | 1    |
| Steel rod 1.5 mm                  | 2    |
| Vent tape                         | 1    |
| Music wire 0.8 mm                 | 2    |
| Wheel collar 3/32 in              | 2    |
| 15T servo arms                    | 1    |
| Brass sheet 1 mm                  | 2    |
| M1.6 nuts                         | 2    |
| XT30 series adapter (2 into 1)    | 1    |



**Motor and prop.** T-Motor F50 2207 2150KV (6S-rated, 24 V; Ø27 x 29.5 mm incl. shaft, 28.56 g, 16 x 16 mm M3 with 4 mm thread depth, 5 mm shaft, M5 x 10.75 mm prop thread, 50 mOhm, 10 1.6 A @10 V; 54.8 A / 1273 W peak): the SkyG Speed 60 motor, same mount. Prop: Gemfan Vortex 5x7.5E fixed electric prop (glass-fibre nylon, 3.7 g, hub 13 x 6 mm, 6.35 mm bore with the 5 mm adapter ring; Robocombo stock). Gemfan publishes no rpm rating: each batch is spin-tested to 45,600 rpm (1.2 x the 38,000 rpm governor ceiling) behind a shield before flight. Pusher: the 5x7.5R, or the E mounted front face toward the motor with the motor reversed in the ESC; rpm-governed.

**rpm governor and current limit.** The flight controller runs skyg\_governor.lua: it holds the prop at or below the stage rpm (never above 38,000 rpm, 92 % of the prop's rating) and the battery current at 60 A, the ESC and motor rating. In level flight the motor reaches about 37,862 rpm at full throttle; the governor acts in dives, at low speed on a full pack and per speed stage. THR\_MAX 70 % is the backstop if the script stops. Prove it on the thrust stand first, and inspect the blades after every thrust-stand run and flying day.

**Battery.** BetaFPV LAVA II 3S 680 95C LiHV (2-pack); XT30; the two packs plug into a 2-into-1 XT30 series adapter. Full throttle in bursts of  $\leq 10$  s, then  $\geq 30$  s at low throttle: the ESC lives on its thermal mass. Static thrust in the model: 1458 g at 60 A; confirm on a thrust stand before the first flight.



# PRINT SETTINGS



The airframe is designed for **Bambu ASA Aero** (foaming, lightweight, heat deflection 85 °C) on a **0.4 mm nozzle** and **0.20 mm layers**; every part fits a Bambu H2S or H2D build volume. The motor plate and spinner are PETG-CF. ASA needs the closed chamber (45–60 °C), a dried spool (80 °C, 8 h) and glue on the plate; print long thin parts (wing halves, ailerons) with a brim against warping.

## GENERAL GUIDELINES

- Fuselage, nose and hatch shells: 2 walls, no infill (the internal ribs and bosses are modelled).
- Wing halves and ailerons: 1 wall, 5 % gyroid.
- Tail panels: 2 walls, 10 % gyroid. Fin: 2 walls, 15 %.
- Motor plate and spinner: PETG-CF, 4 walls, 40 %.
- Tall standing parts: 5 mm brim, slow outer wall, enclosure door closed.

## SUPPORTS: OFF

Every part is designed to print without supports in the orientation shown (DFAM rules, print audit: all parts pass). Keep supports switched off. The only bridge longer than 12 mm is under the rear wing-screw nut in F2, hidden inside the wing bay: if it sags on your printer, paint a small support under it there. Check the top of the aileron's torque-rod bore on the first aileron.

# PRINT SETTINGS

## FIRST PART AND WEIGHT CHECK

Print F1 first and weigh it. If it is more than 15 % over the sliced mass (about 12 g), lower the ASA Aero temperature or flow before printing the rest; foaming filaments vary with temperature, humidity and the enclosure.

## COUPON TESTS

Before the full build, print the last 25 mm of F1 with the first 25 mm of F2, and the F2/F3 and nose joints the same way: every joint must slide on and off by hand and the M2 screws must bite. Print a 40 mm wing-root section and check the spar fit.

## SLICER

The masses in this manual come from Bambu Studio with the H2S 0.4 mm profile and the walls/infill listed on the orientation pages. Import the STL files in millimetres; keep each part in the orientation shown.

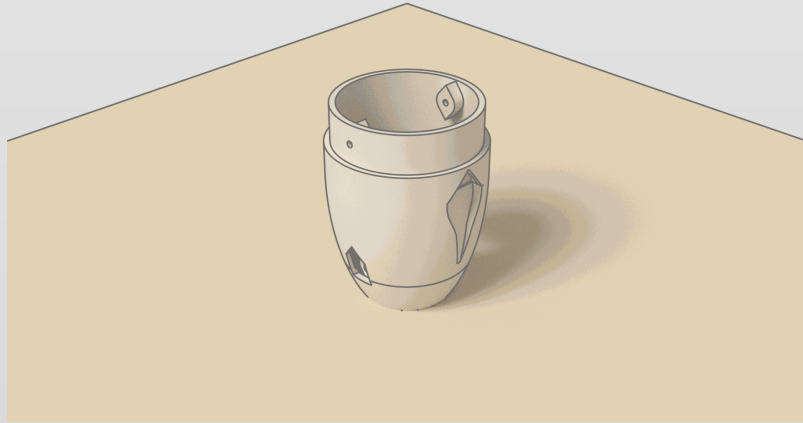
# PARTS LIST

| PART                      | MATERIAL | MASS   |
|---------------------------|----------|--------|
| N0M NOSE (MICRO FPV)      | ASA Aero | 4.3 g  |
| N0P NOSE (PI CAMERA)      | ASA Aero | 4.0 g  |
| N0MD NOSE (MICRO + DOME)  | ASA Aero | 2.7 g  |
| N0PD NOSE (PI + DOME)     | ASA Aero | 2.9 g  |
| N0O NOSE (DJI O4)         | ASA Aero | 5.3 g  |
| N0OD NOSE (DJI O4 + DOME) | ASA Aero | 4.3 g  |
| F1 FUSELAGE FRONT         | ASA Aero | 12.2 g |
| F2 FUSELAGE MAIN          | ASA Aero | 14.6 g |
| F3 TAIL MODULE            | ASA Aero | 7.2 g  |
| HF FRONT HATCH            | ASA Aero | 2.7 g  |
| FN FIN                    | ASA Aero | 3.7 g  |
| VTX SHELF (N0M)           | ASA Aero | 0.3 g  |
| DOME RING                 | ASA Aero | 0.3 g  |
| HG GPS CARRIER            | ASA Aero | 0.8 g  |

| PART                | MATERIAL | MASS   |
|---------------------|----------|--------|
| WR WING HALF RIGHT  | ASA Aero | 10.9 g |
| WL WING HALF LEFT   | ASA Aero | 11.5 g |
| AR AILERON RIGHT    | ASA Aero | 1.5 g  |
| AL AILERON LEFT     | ASA Aero | 1.3 g  |
| SR TAIL PANEL RIGHT | ASA Aero | 2.2 g  |
| SL TAIL PANEL LEFT  | ASA Aero | 2.1 g  |
| M1 MOTOR PLATE      | PETG-CF  | 1.9 g  |
| SP SPINNER          | PETG-CF  | 1.7 g  |

One aircraft with the N0M nose: 79 g printed. Only one camera nose is needed; the others are options.

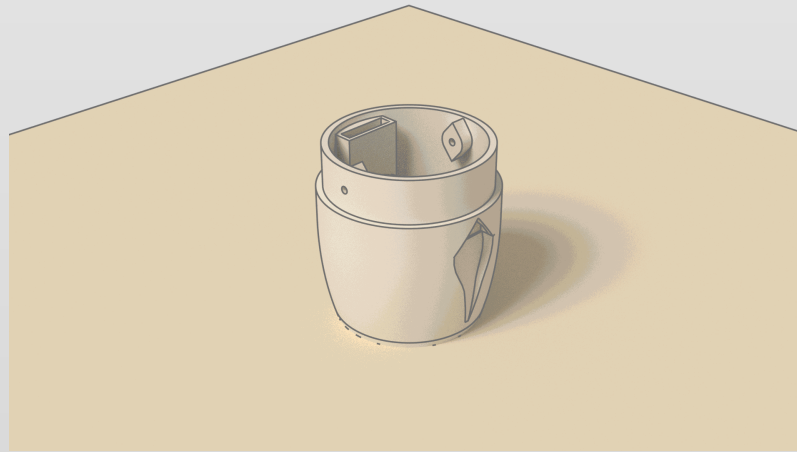
# PARTS ORIENTATION



## **N0M NOSE (MICRO FPV)**

FACE DOWN, SPIGOT UP

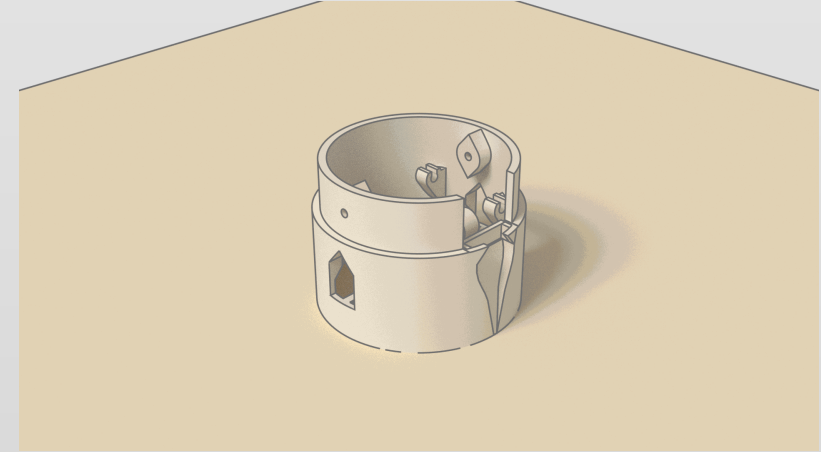
2 WALLS · 0 % INFILL · ASA AERO



## **N0P NOSE (PI CAMERA)**

FACE DOWN, SPIGOT UP

2 WALLS · 0 % INFILL · ASA AERO

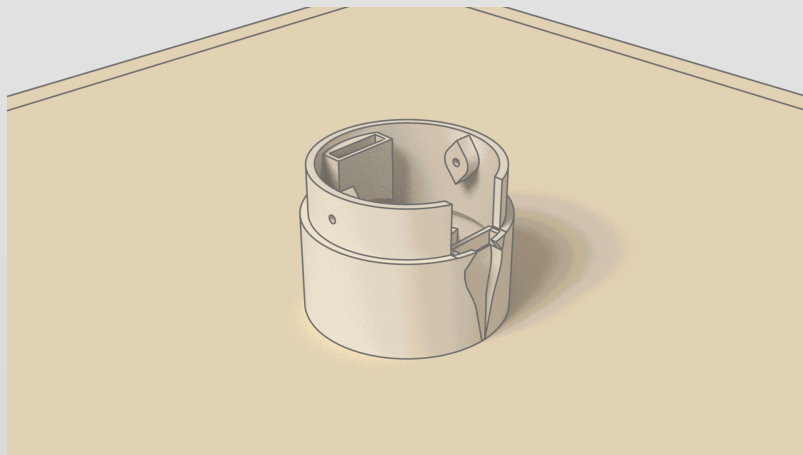


## **N0MD NOSE (MICRO + DOME)**

FRONT FACE DOWN

2 WALLS · 0 % INFILL · ASA AERO

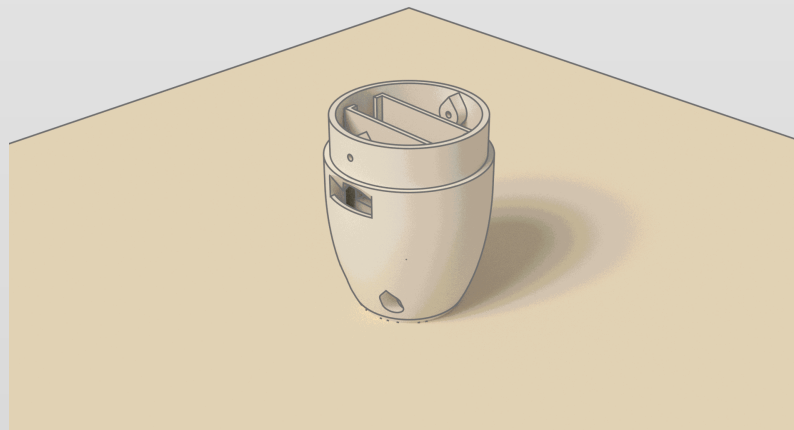
# PARTS ORIENTATION



## **N0PD NOSE (PI + DOME)**

FRONT FACE DOWN

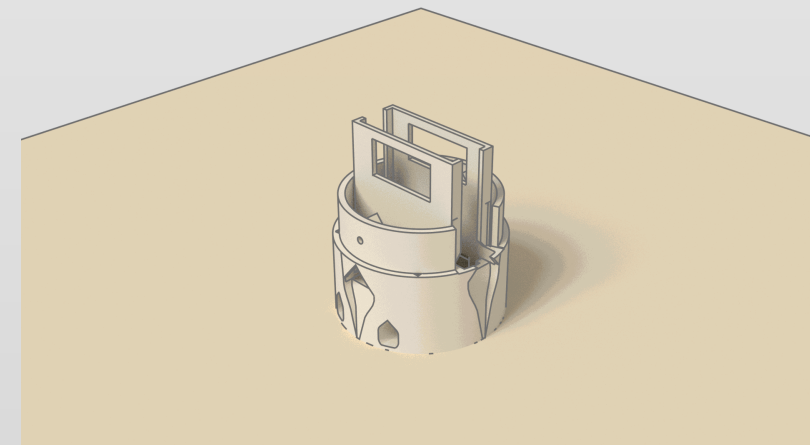
2 WALLS · 0 % INFILL · ASA AERO



## **N00 NOSE (DJI O4)**

FACE DOWN

2 WALLS · 0 % INFILL · ASA AERO

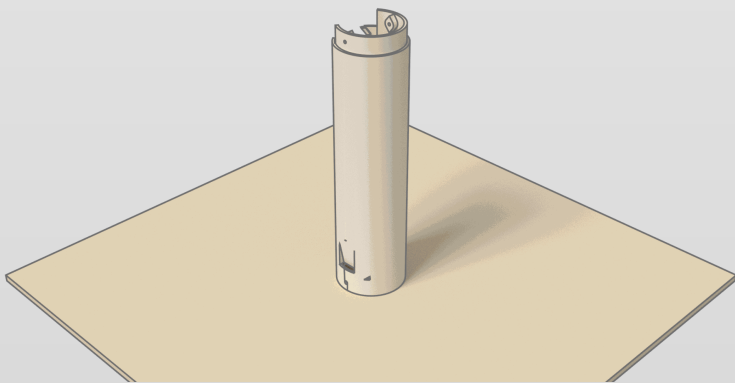


## **N00D NOSE (DJI O4 + DOME)**

FRONT FACE DOWN

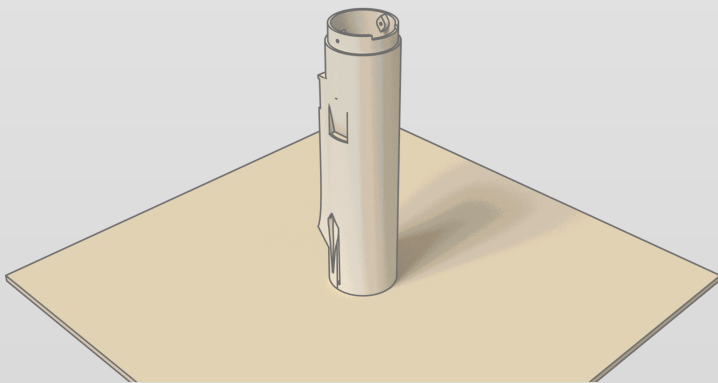
2 WALLS · 0 % INFILL · ASA AERO

# PARTS ORIENTATION



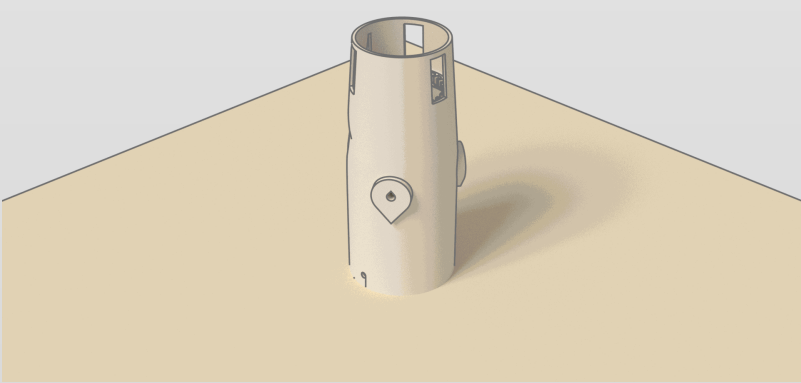
## F1 FUSELAGE FRONT

FRONT END DOWN · SPIGOT UP · 5 mm BRIM  
2 WALLS · 0 % INFILL · ASA AERO



## F2 FUSELAGE MAIN

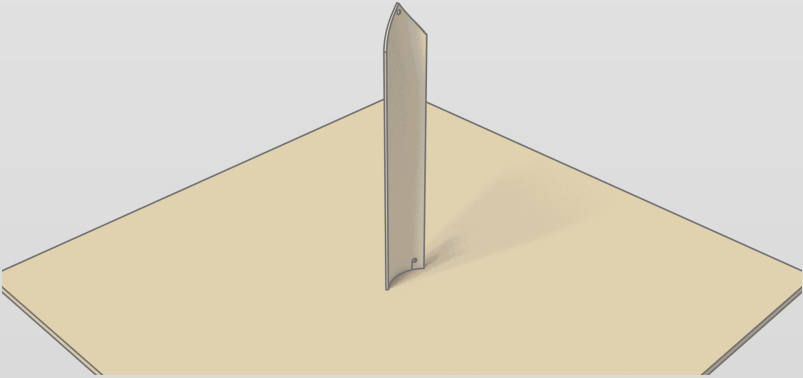
FRONT END DOWN · SPIGOT UP · 5 mm BRIM  
2 WALLS · 0 % INFILL · ASA AERO



## F3 TAIL MODULE

FRONT END (SOCKET) DOWN · 5 mm BRIM  
2 WALLS · 0 % INFILL · ASA AERO

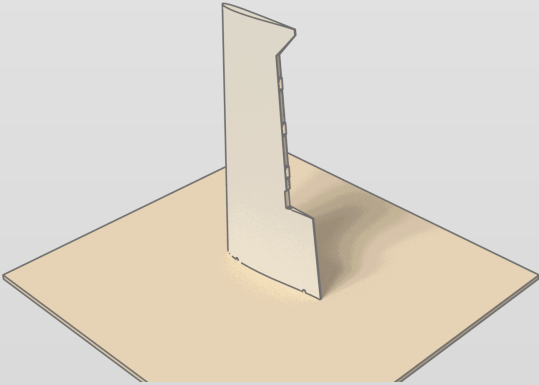
# PARTS ORIENTATION



**HF FRONT HATCH**

STANDING, FRONT END DOWN

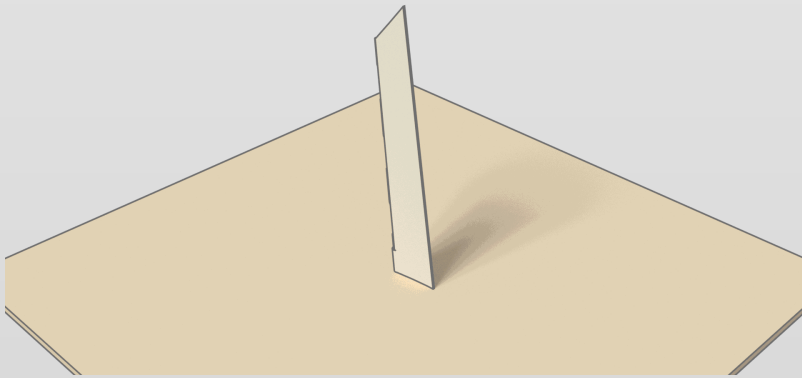
2 WALLS · 0 % INFILL · ASA AERO



**WR WING HALF R/L**

ON THE ROOT RIB

1 WALLS · 5 % INFILL · ASA AERO

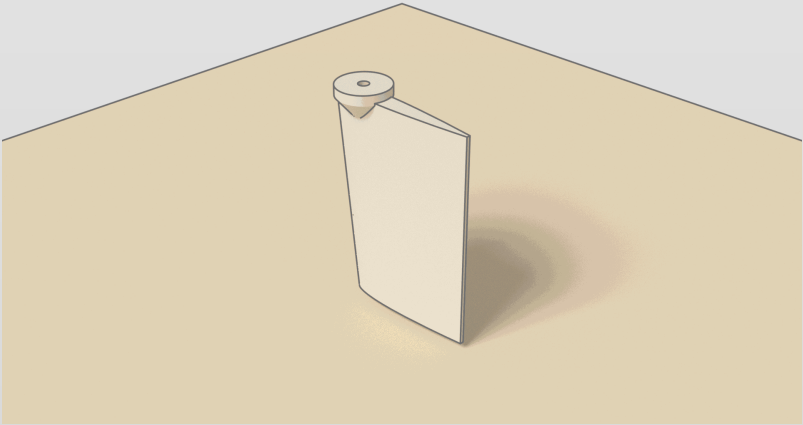


**AR AILERON R/L**

ON THE INBOARD END

1 WALLS · 5 % INFILL · ASA AERO

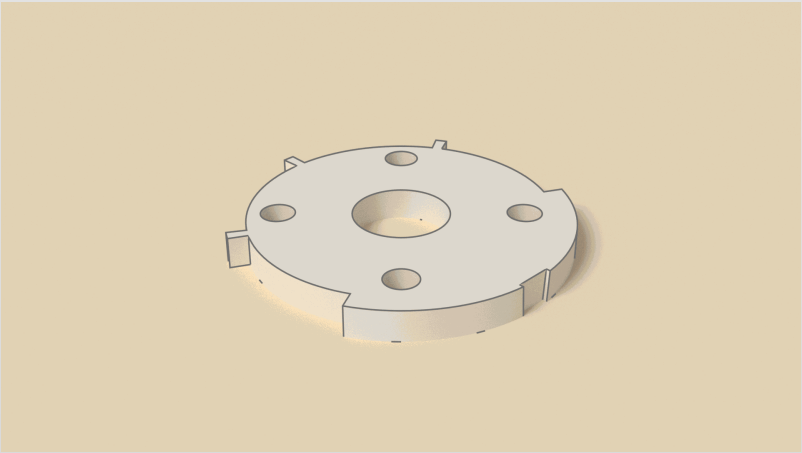
# PARTS ORIENTATION



**SR TAIL PANEL R / L**

TIP DOWN, HUB UP

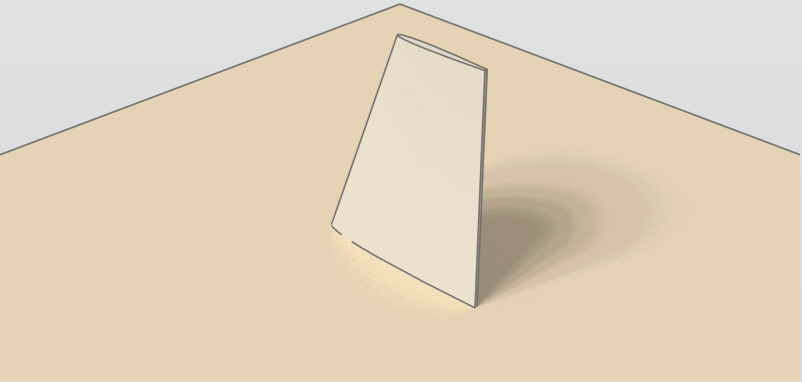
2 WALLS · 10 % INFILL · ASA AERO



**M1 MOTOR PLATE**

FLAT

4 WALLS · 40 % INFILL · PETG-CF

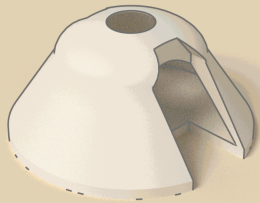


**FN FIN**

ON THE ROOT TAB

2 WALLS · 15 % INFILL · ASA AERO

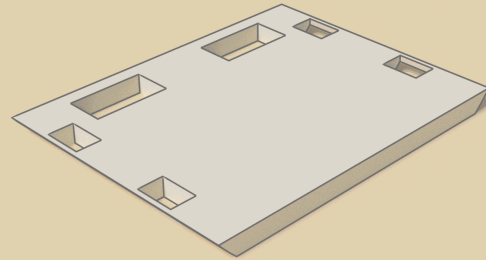
# PARTS ORIENTATION



## SP SPINNER

BACKPLATE DOWN

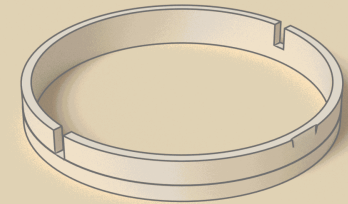
4 WALLS · 40 % INFILL · PETG-CF



## VTX SHELF (NOM)

FLAT

2 WALLS · 15 % INFILL · ASA AERO

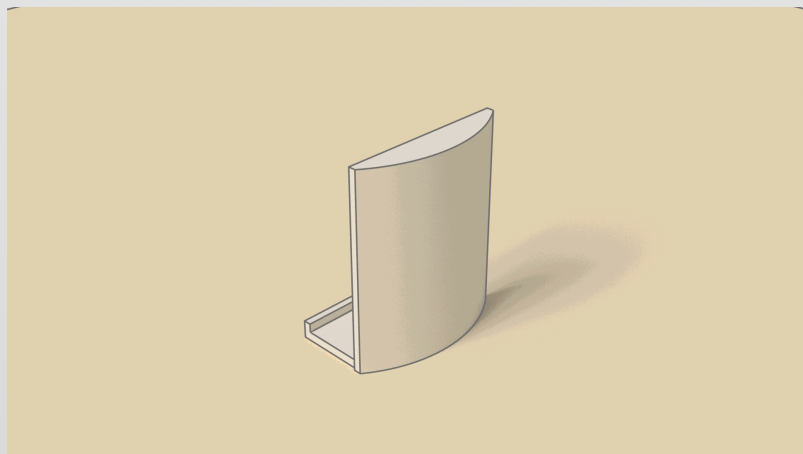


## DOMERING

LIP DOWN

2 WALLS · 15 % INFILL · ASA AERO

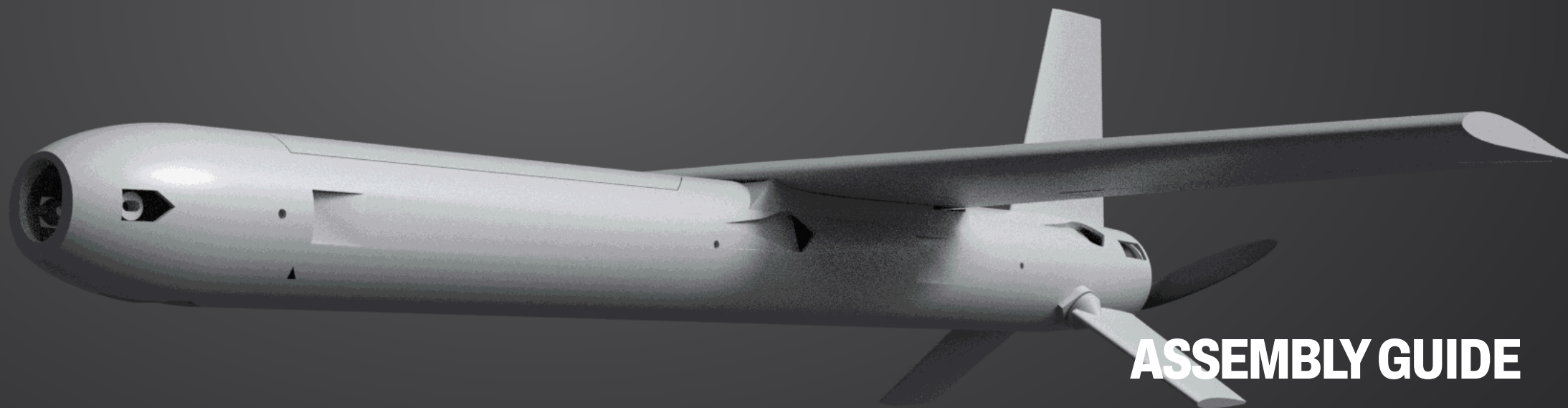
# PARTS ORIENTATION



## HG GPS CARRIER

ON ITS REAR WALL

2 WALLS · 15 % INFILL · ASA AERO

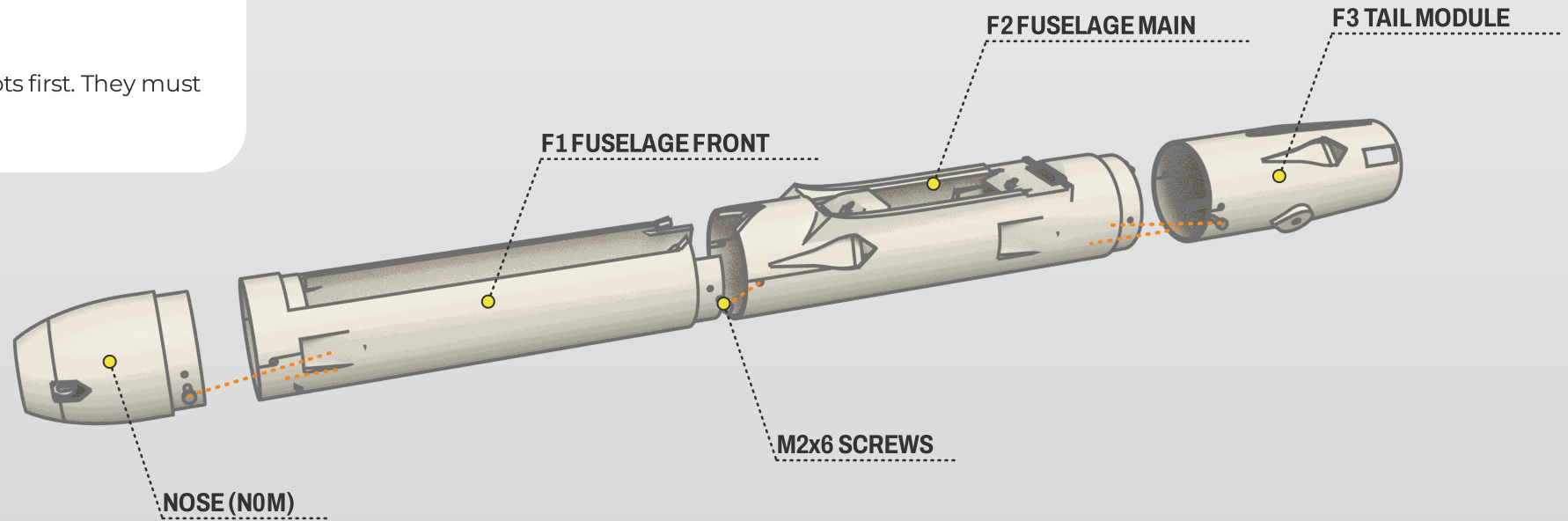


# FUSELAGE ASSEMBLY



The fuselage is four modules on spigot joints: the camera nose, F1, F2 and the F3 tail module. Each joint is held by two M2 × 6 screws from the sides into bosses inside the spigot; nothing is glued, so every module comes off for service.

Clean any stringing from the spigots first. They must slide in by hand.

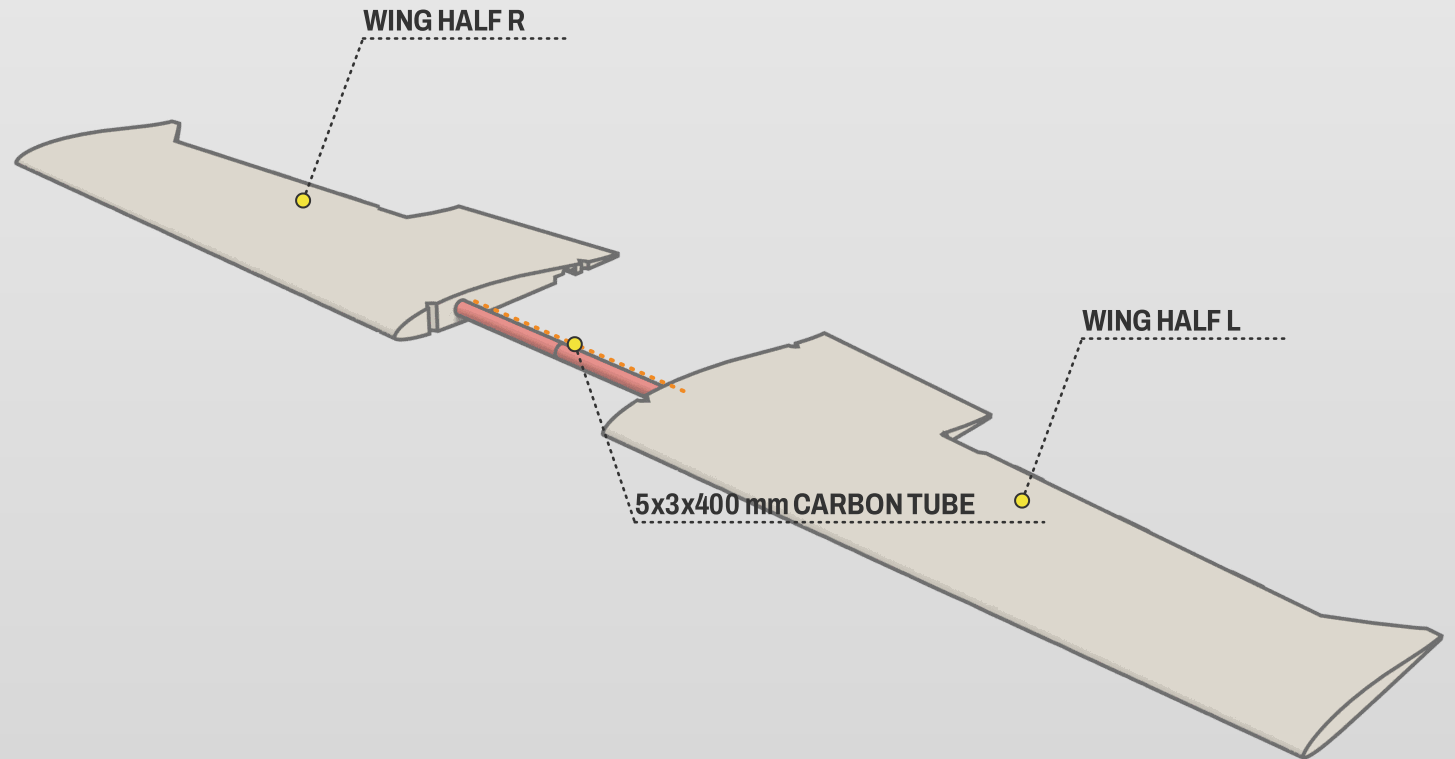


M2 × 6 SCREWS

# WING ASSEMBLY



Cut one 400 mm length of 5 × 3 mm carbon tube and slide it through both wing halves (200 mm into each). Epoxy the tube in the bores and the two root ribs together with the whole wing flat on a table (0° dihedral).

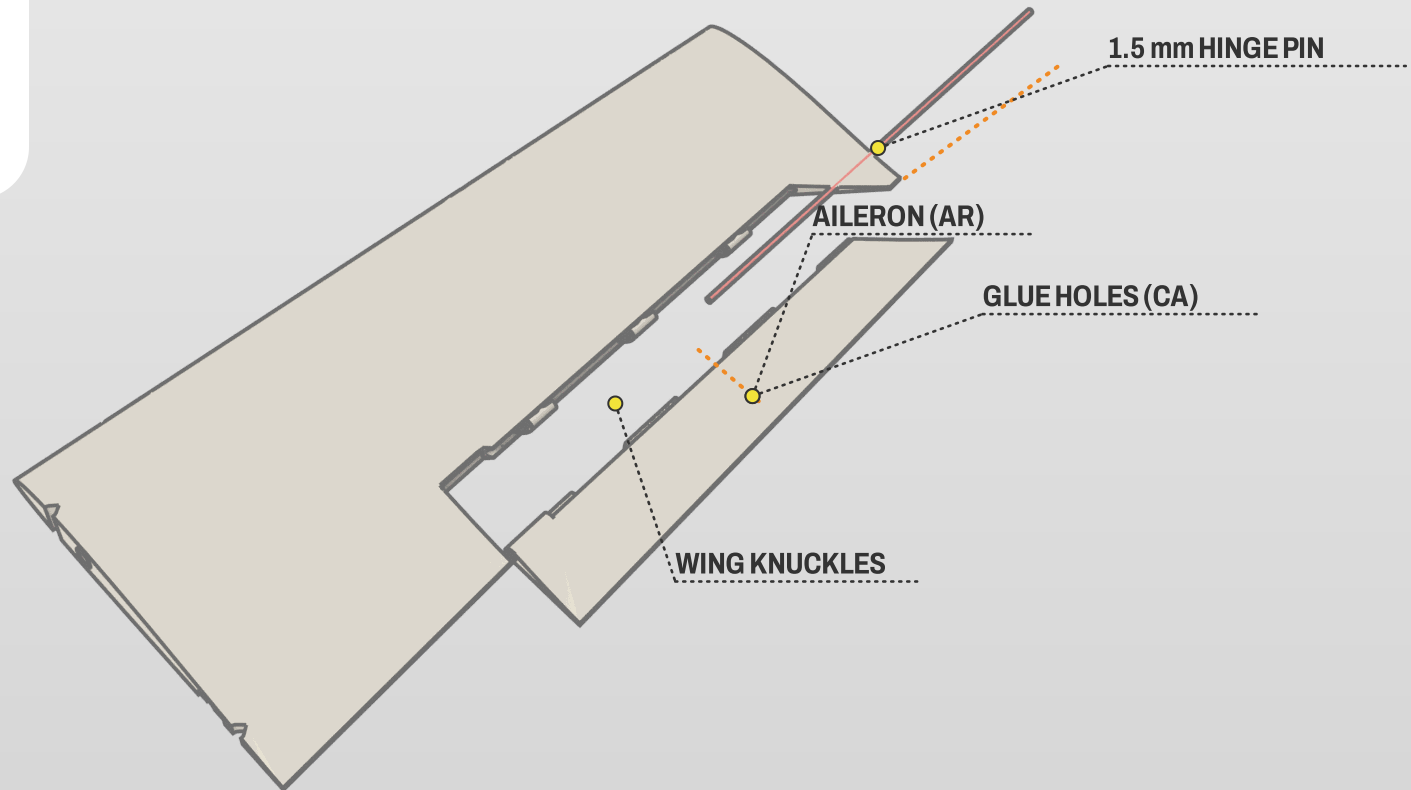


EPOXY

# WING ASSEMBLY



Set each aileron in place: its rounded nose sits in the cove along the wing's hinge face and the wing's three knuckles enter the pockets in the nose. Push a 132 mm length of 1.5 mm steel rod in from the wingtip through the tip, the aileron and the knuckles. Wick thin CA into the aileron's four glue holes only, then plug the tip hole. The aileron must swing freely.

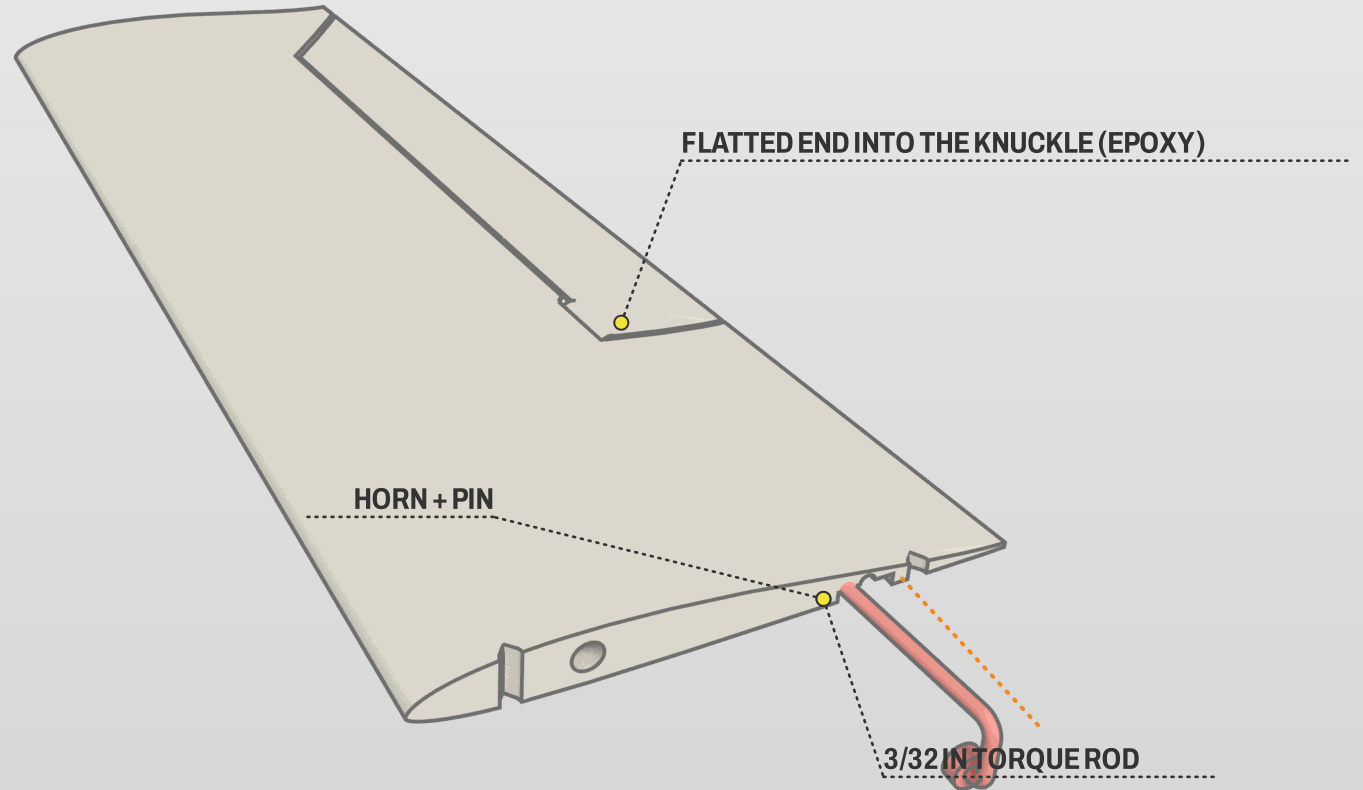


THIN CA

# WING ASSEMBLY



Bend a 12 mm horn down at one end of each 3/32 in (2.38 mm) rod with a 5.5 mm pin outboard at its tip, and file a 0.3 mm flat on the last 9 mm of the straight end, opposite the horn. Push it in from the root face until the flat enters the D-bore of the aileron's root knuckle; epoxy the end and hold the aileron 3° trailing edge up while it cures.

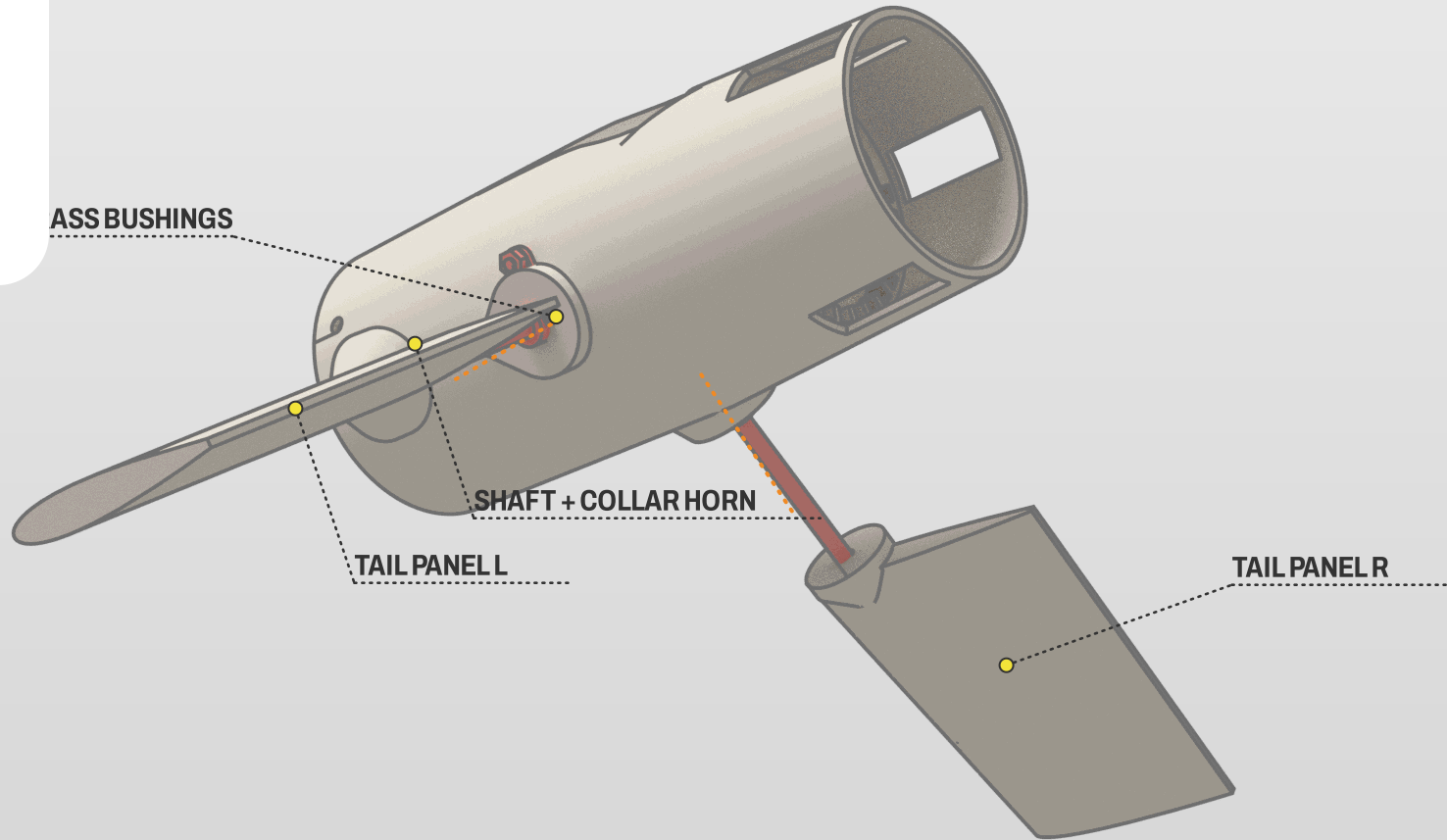


EPOXY

# TAIL MODULE



Press a 10 mm brass bushing into each side boss of F3. Epoxy a 2.5 mm steel shaft 35 mm into each tail panel's root bore. From the open front of F3, push each shaft through its bushing and fit the wheel-collar horn with its 8 mm brass plate standing along the panel's normal (not pointing forward), 0.5 mm end play, set screw on a filed flat.

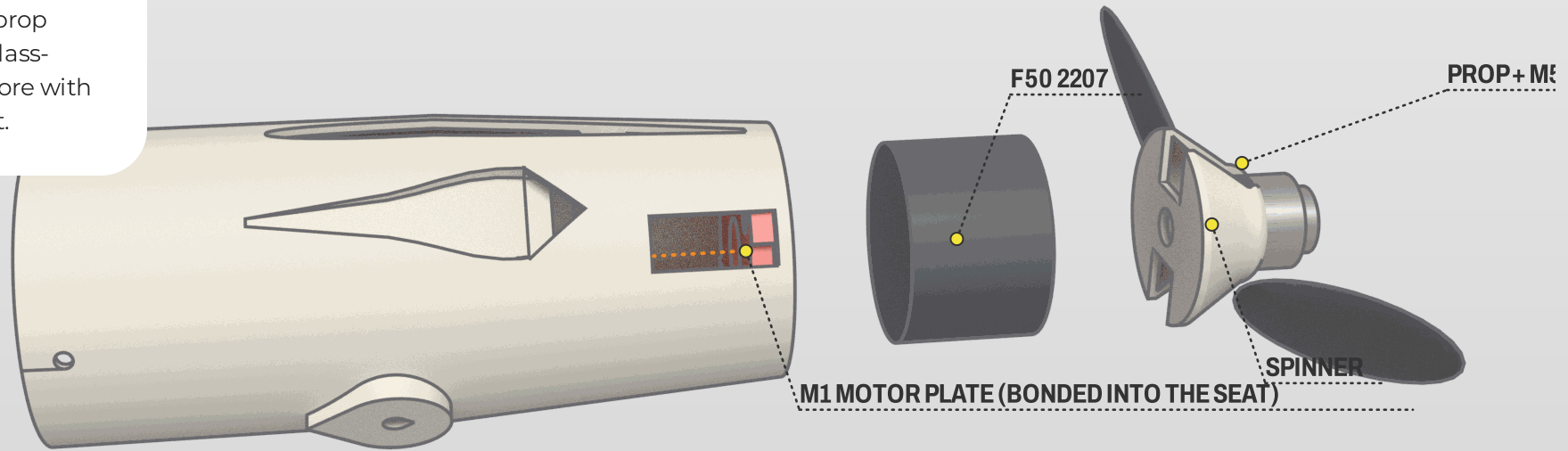


EPOXY

# TAIL MODULE



Bond the M1 plate into the conical seat at the rear of F3 with high-temperature epoxy, its edge notches lined up with the cooling windows; it is the motor bulkhead. Then screw the motor to it with its 4 × M3 from inside the module, through the 28 mm opening. Do this before the tail-panel shafts and servos go in. Fit the spinner backplate, the prop (Gemfan Vortex 5x7.5E fixed electric prop (glass-fibre nylon, 3.7 g, hub 13 x 6 mm, 6.35 mm bore with the 5 mm adapter ring) and the thin M5 nut.

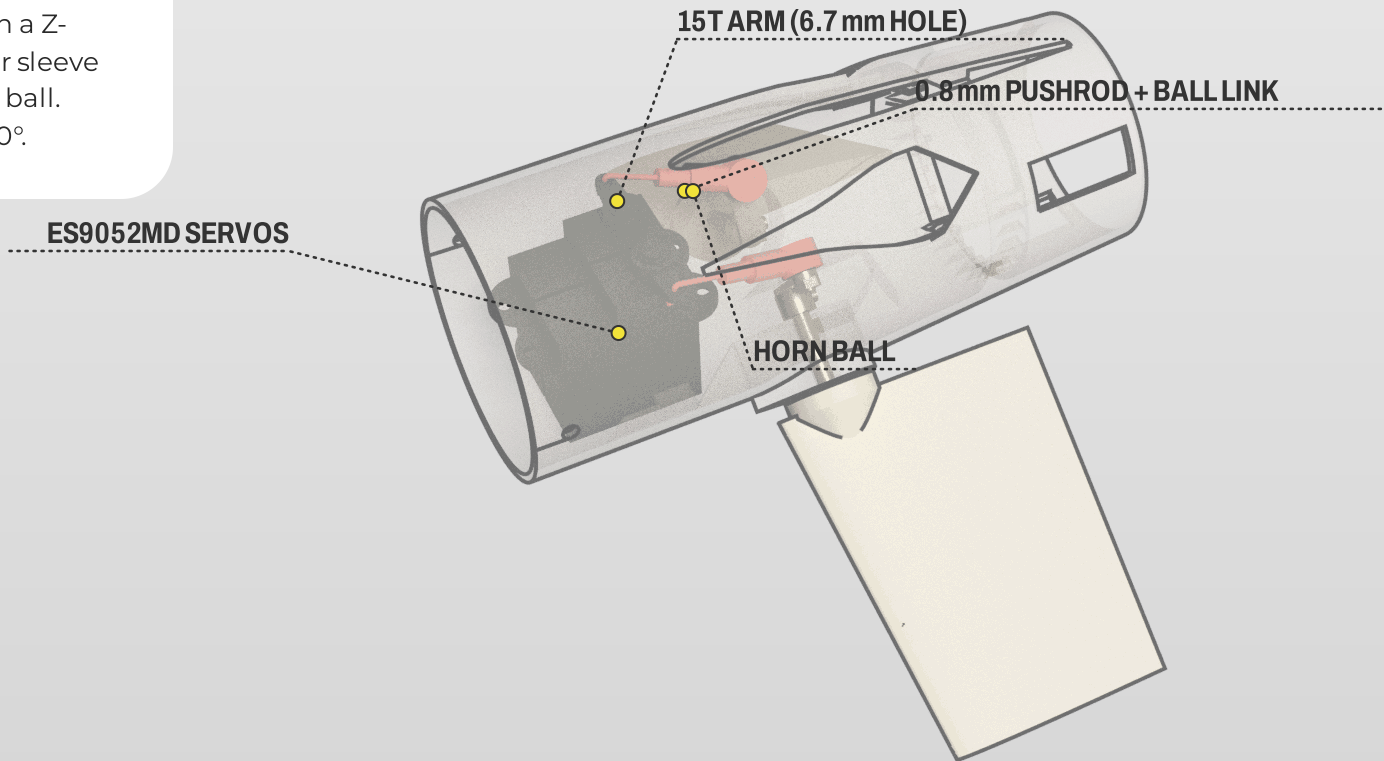


EPOXY

# TAIL MODULE



Stand both ES9052MD servos through the cut-outs in the F3 floor on foam tape on the bottom skin, a strap over them, output at the aft end, leads along the side walls. One-sided 15T arm pointing outboard; 0.8 mm music-wire pushrod with a Z-bend in the arm's 6.7 mm hole and a solder sleeve into the micro ball link on the horn's 3 mm ball. Check both panels move freely through  $\pm 20^\circ$ .

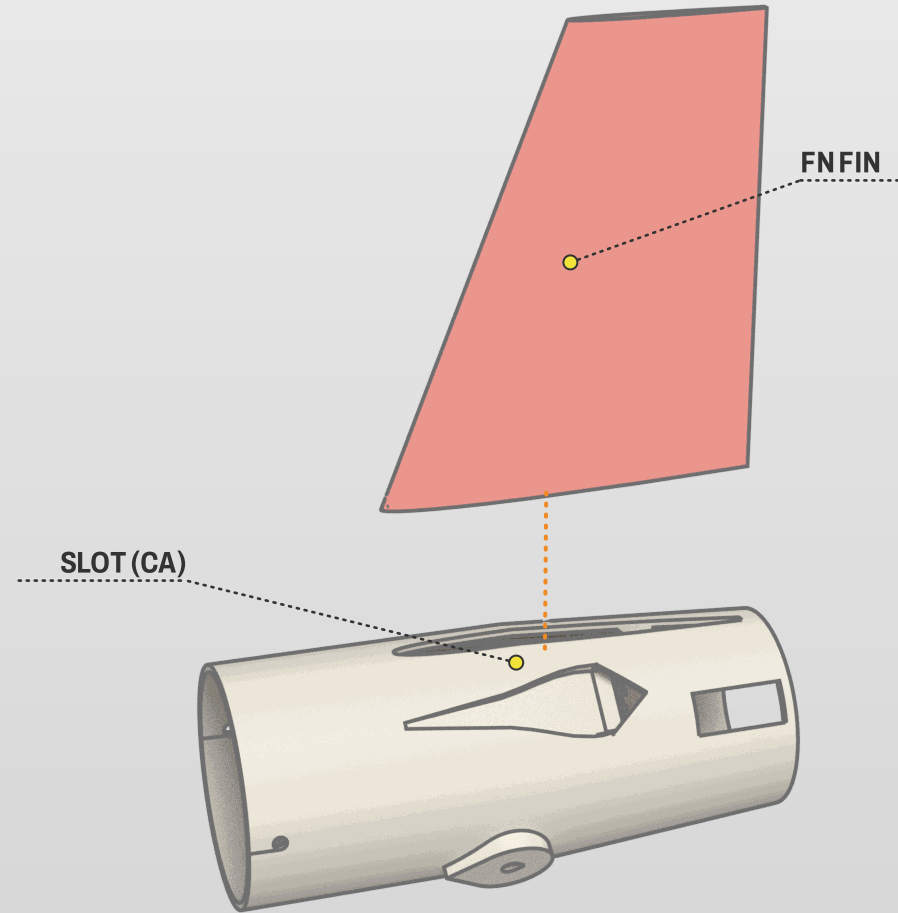


SERVO TAPE

# TAIL MODULE



Glue the fin into the top slot of F3 with CA, vertical and square to the wing.

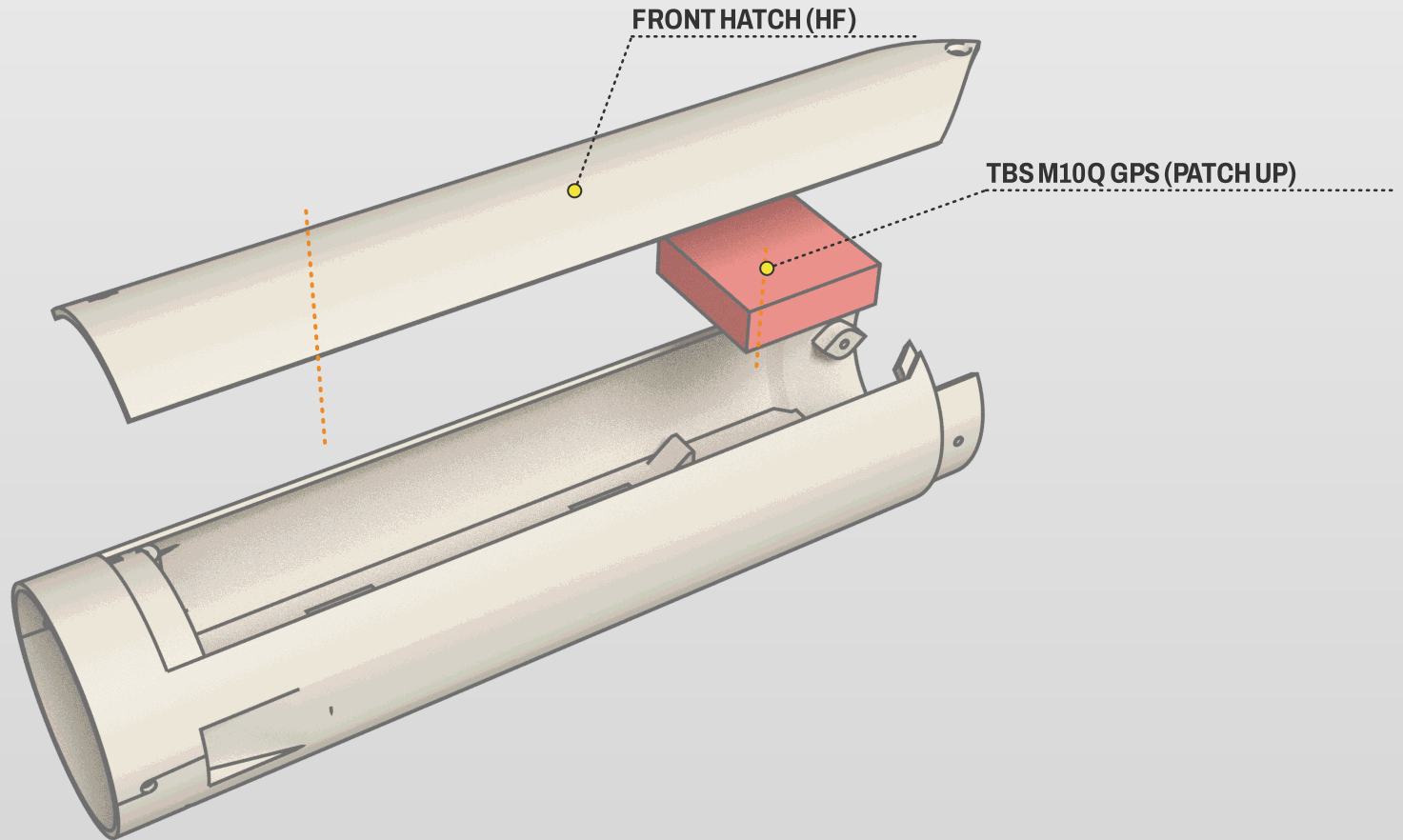


CA GLUE

# FUSELAGE ASSEMBLY



Bond the printed GPS carrier (HG) under the rear of the front hatch skin. Foam-tape the TBS M10Q GPS, patch up, to its ceiling, against its rear ledge. The hatch's pointed rear end reaches into F2 and its rear screw goes into F2: take the hatch off before separating F1 and F2. Close the hatch with the two M2 × 6 countersunk screws at its ends (they self-tap into the bosses under the lips; do not overtighten). Leave a slack loop in the GPS lead so the hatch lifts off.

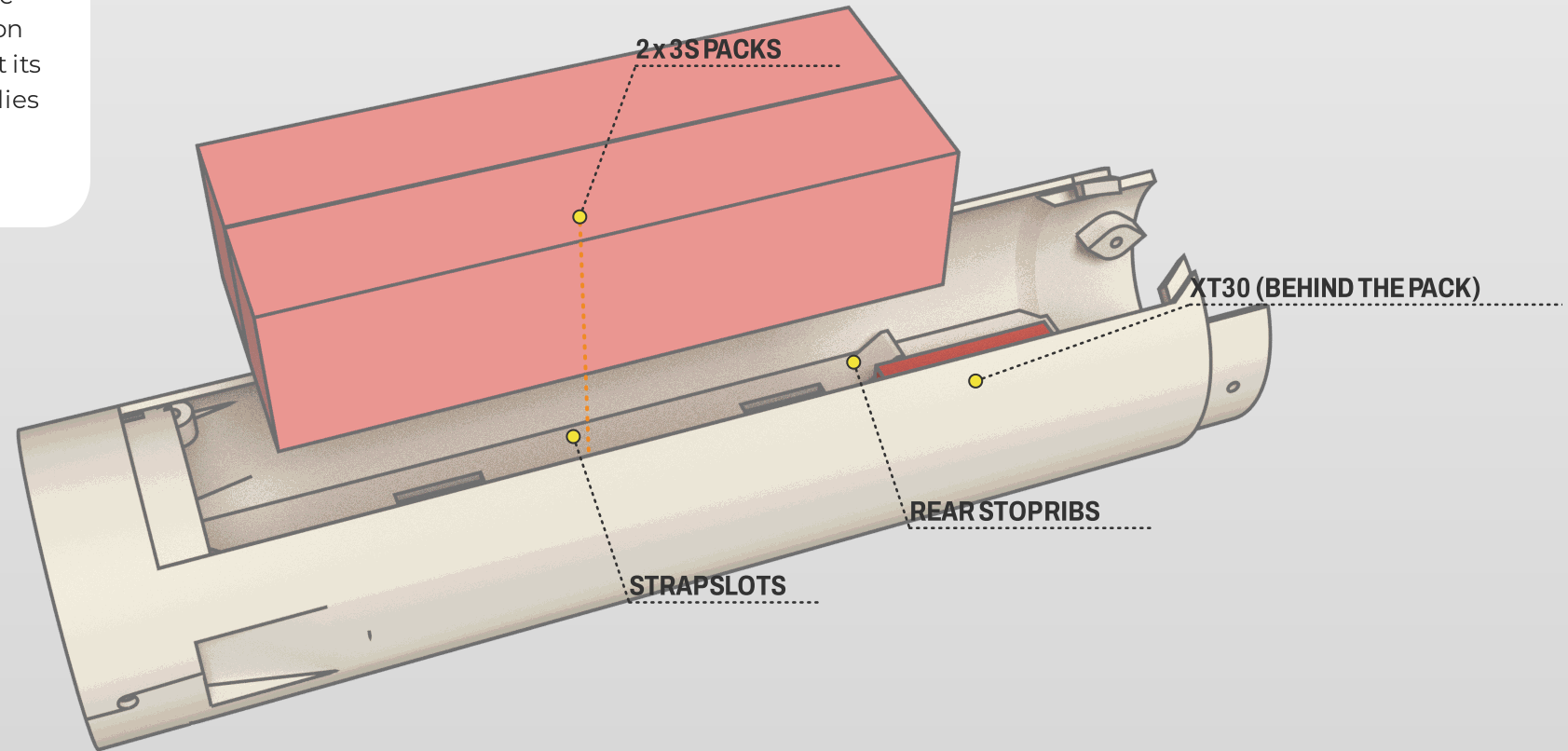


M2 × 6 SCREWS

# FUSELAGE ASSEMBLY



The two 3S packs sit side by side on their sides on the F1 floor, joined by the XT30 series adapter; one strap through the slots holds both. The pack sits on the F1 floor with a strap through the slots, leads at its rear. Two floor ribs are its rear stop; the XT30 pair lies behind the pack under the GPS and unplugs through the front hatch.

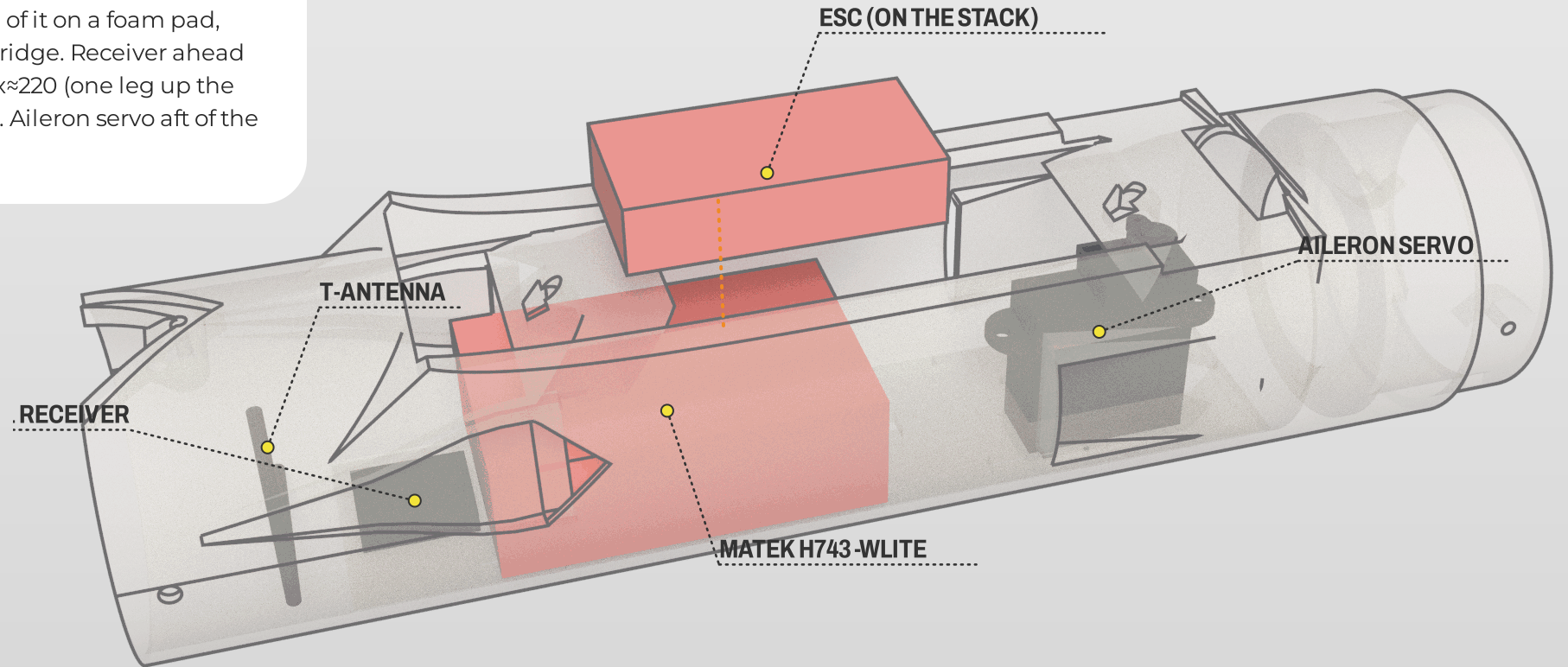


VELCRO STRAP

# EQUIPMENT



Matek H743-WLITE on a 0.5 mm foam pad and a strap on the wing-bay floor, arrow forward, servo rail (Vx) set to 6 V; the ESC on top of it on a foam pad, behind the front wing-bolt bridge. Receiver ahead of it, its T-antenna as an L at  $x \approx 220$  (one leg up the left wall, one across the floor). Aileron servo aft of the stack, output at the aft end.

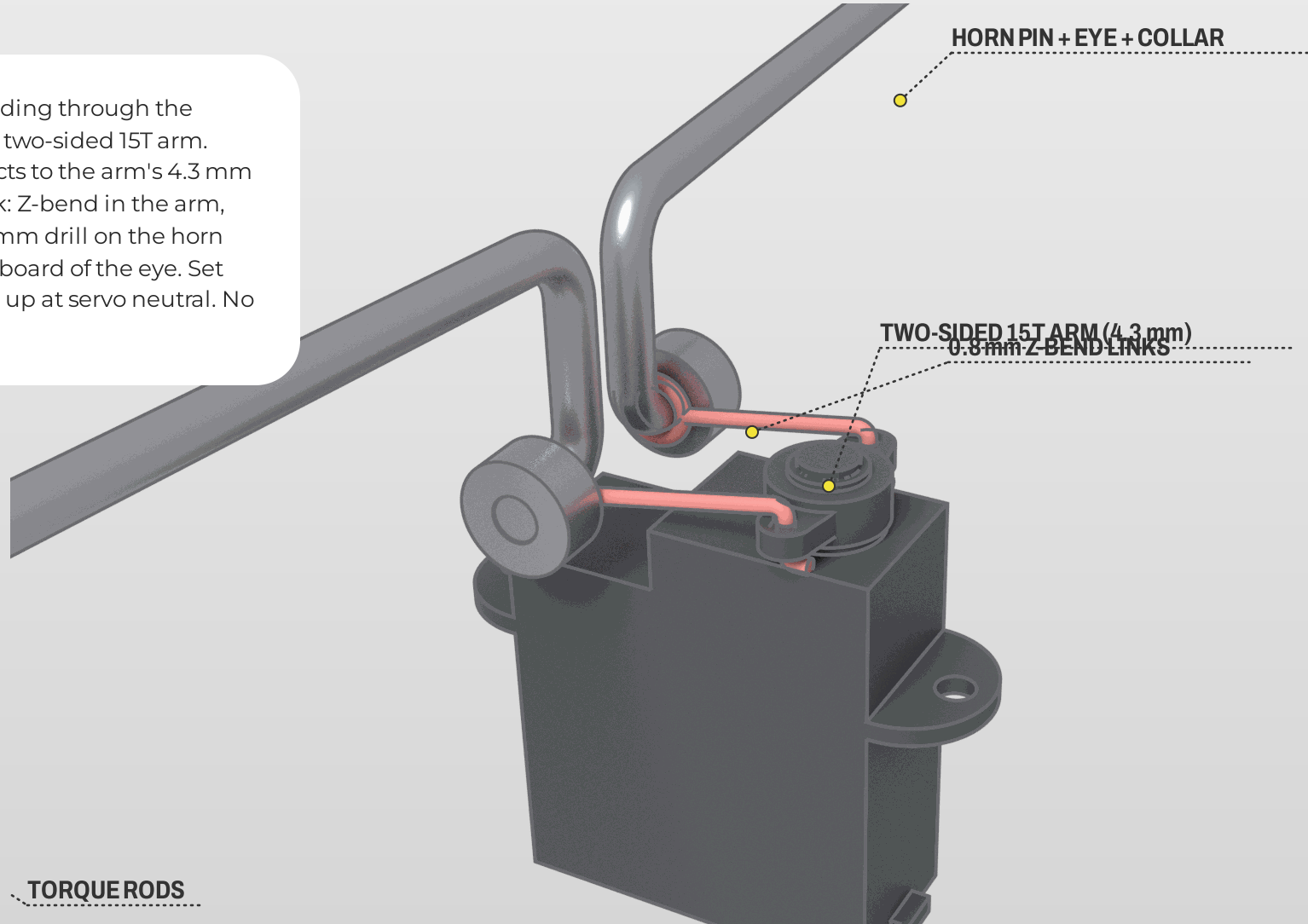


FOAM TAPE

# EQUIPMENT



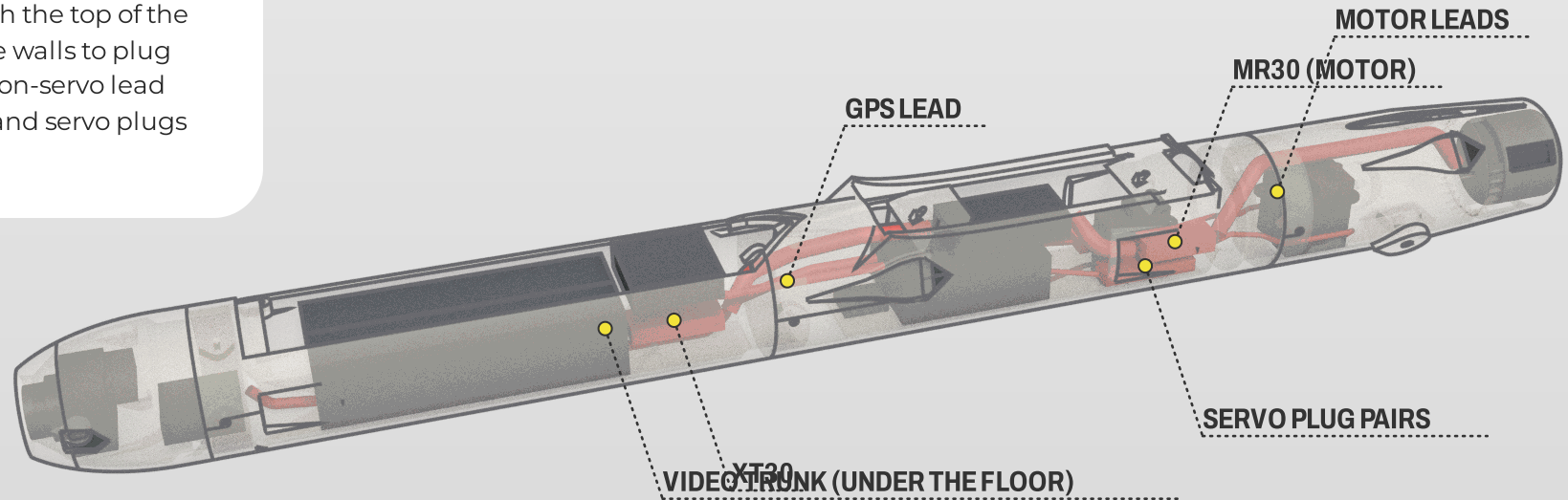
ES9052MD aileron servo standing through the wing-bay floor on foam tape; two-sided 15T arm. Each torque-rod horn connects to the arm's 4.3 mm hole with its own 0.8 mm link: Z-bend in the arm, an eye wrapped round a 2.4 mm drill on the horn pin, a 3/32 in wheel collar outboard of the eye. Set both ailerons 3° trailing edge up at servo neutral. No play at either end.



# EQUIPMENT



Route the leads as shown: video cable under the battery floor and up through the floor hole ahead of the F1/F2 bulkhead; ESC battery lead up the right side over the FC; motor leads low on the left of the wing bay to the MR30, then through the top of the joint; tail-servo leads along the side walls to plug pairs beside the aileron servo; aileron-servo lead under the floor. Unplug the MR30 and servo plugs before pulling the tail module.

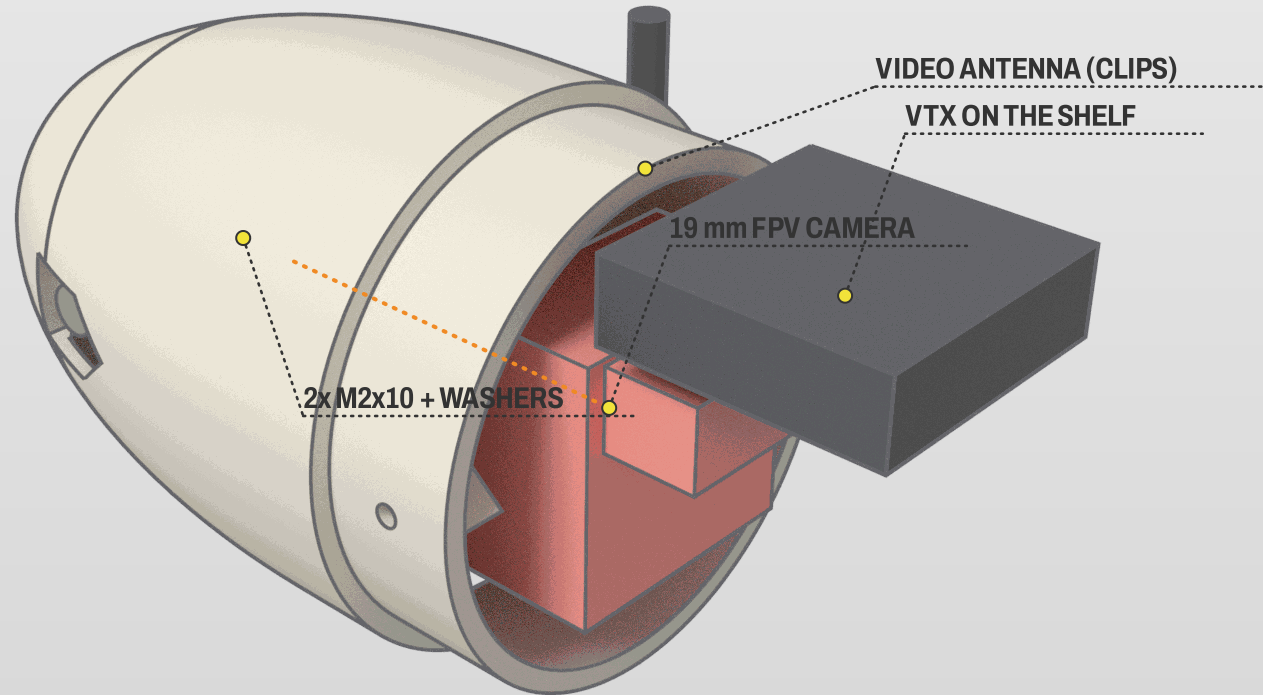


HEAT SHRINK

# CAMERA NOSE



Slide the 19 mm camera in from the open rear until the lens reaches the face; 2 × M2 × 8 pan head + 2 × M2 washer (5 mm OD) through the side seats, lens flush, tilt 0–10° up. Fix the VTX to the printed VTX shelf (foam tape + zip tie through its slots), slide the shelf in from the rear onto its two rails and glue it; video antenna in the two clips on the right wall. Plug the one lead at the nose joint. Dome noses: glue the printed dome ring 3 mm into the front of the tube, then fit the clear dome over its lip with canopy glue (no CA).

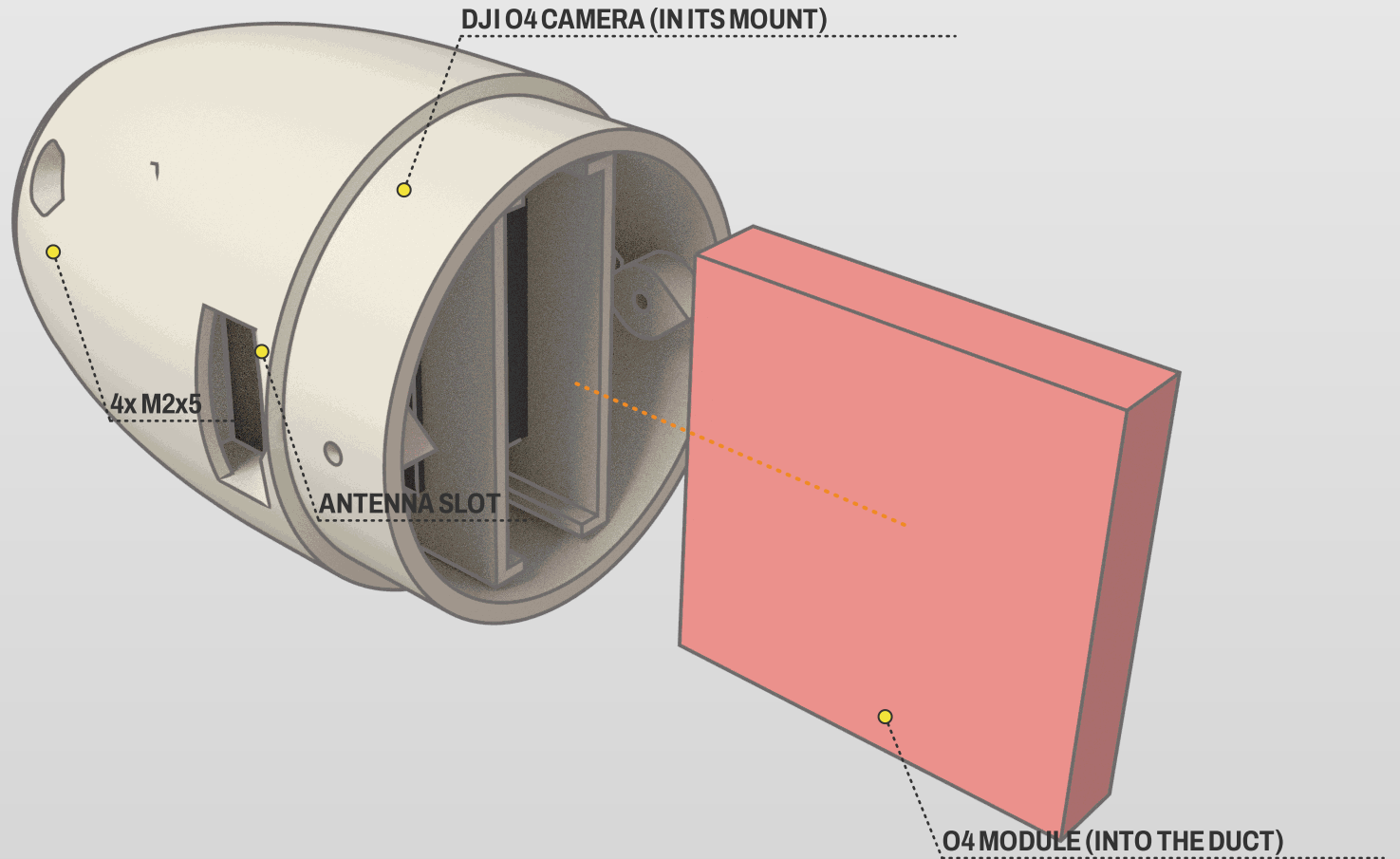


FOAM TAPE

# CAMERA NOSE



DJI O4 option: clamp the camera in its mount with 4 × M2 × 5 through the side slots (at most 3.1 mm into the mount). Plug the coax and 3-in-1 lead into the module and slide it edge-first into the duct until it stops; foam strip in the top groove, tape over the rear edge, never over its faces.

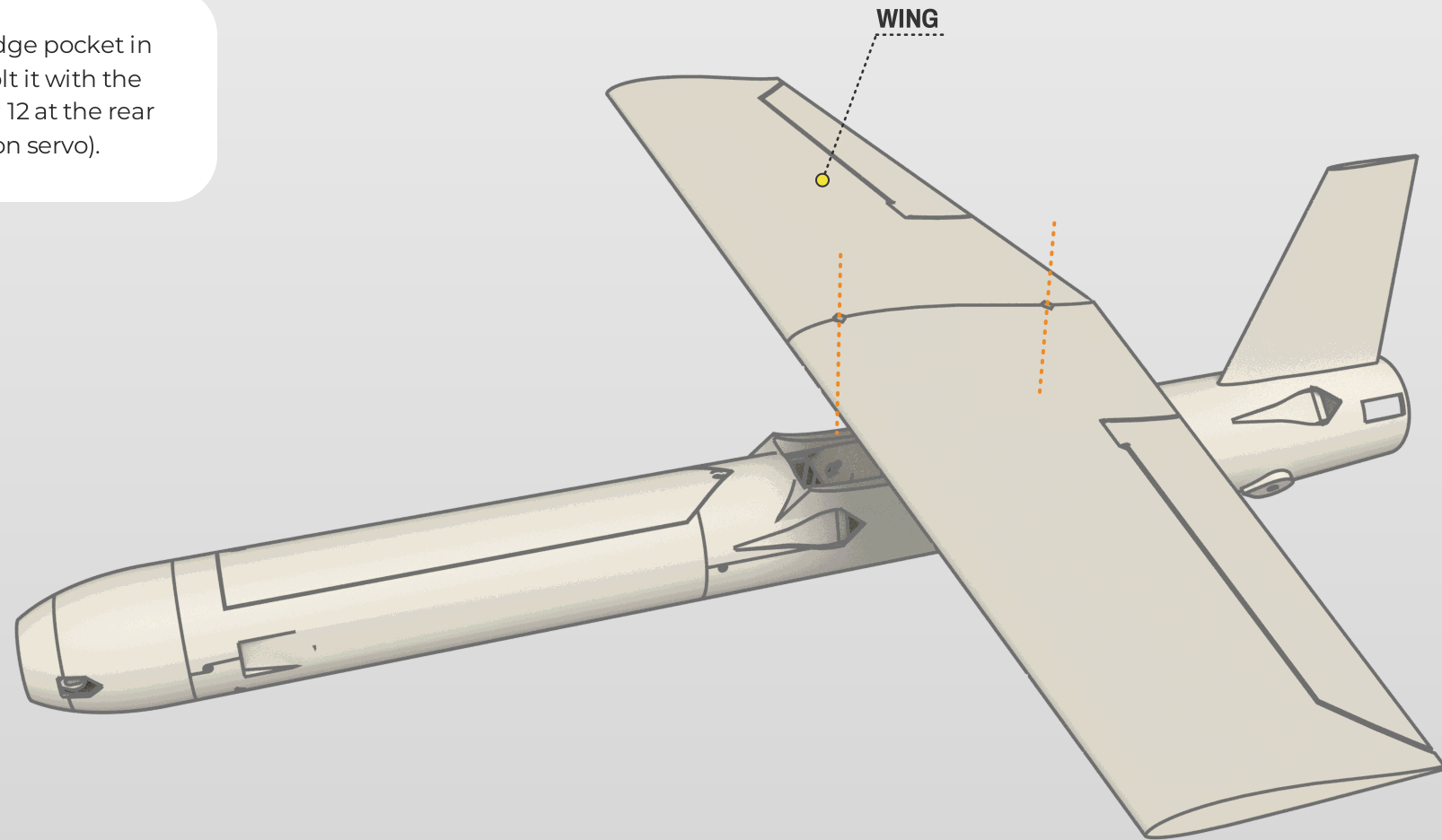


M2 × 5 SCREWS

# FINAL ASSEMBLY



Push an M3 nut up into each bolt-bridge pocket in F2. Set the wing on the saddle and bolt it with the nylon screws: M3 × 16 at the front, M3 × 12 at the rear (a longer rear screw reaches the aileron servo).



NYLON SCREWS

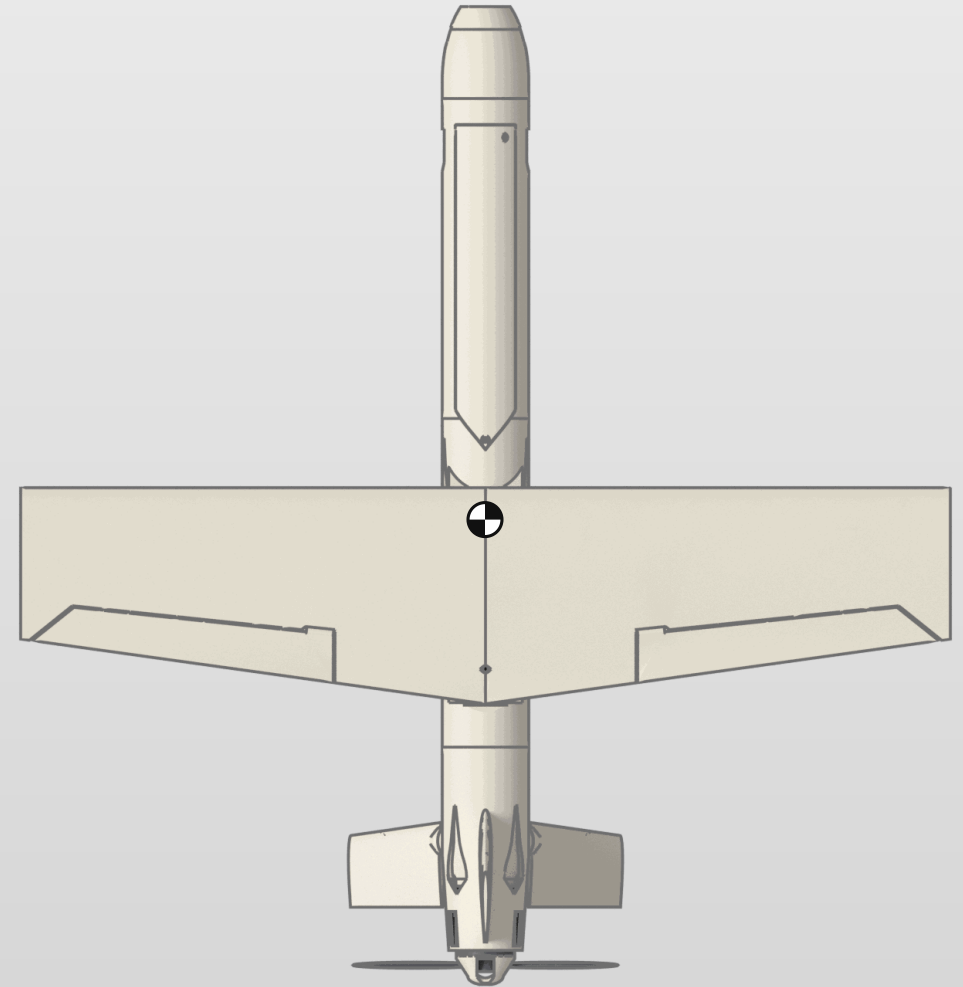
# FINISHING BUILD



Before the first flight: every control surface moves the right way (pitch up: both panel trailing edges up; right roll: right aileron up), throws  $\pm 10^\circ$  on the tail panels, about  $\pm 10^\circ$  ( $\pm 3.5$  mm at the trailing edge) on the ailerons, 30 % expo; the prop turns the right way and pushes aft; failsafe set to motor off.

**Balance:** CG 15 mm behind the wing leading edge at the root (mark it on the wing underside). Slide the battery to trim; for the first flight set the CG 2 mm forward. If a heavy camera needs it, use the nose or tail ballast cup.

**Glass skin:** two  $\pm 45^\circ$  layers of 49 g/m<sup>2</sup> glass on the wing halves and tail panels after printing, then a thin paint finish.

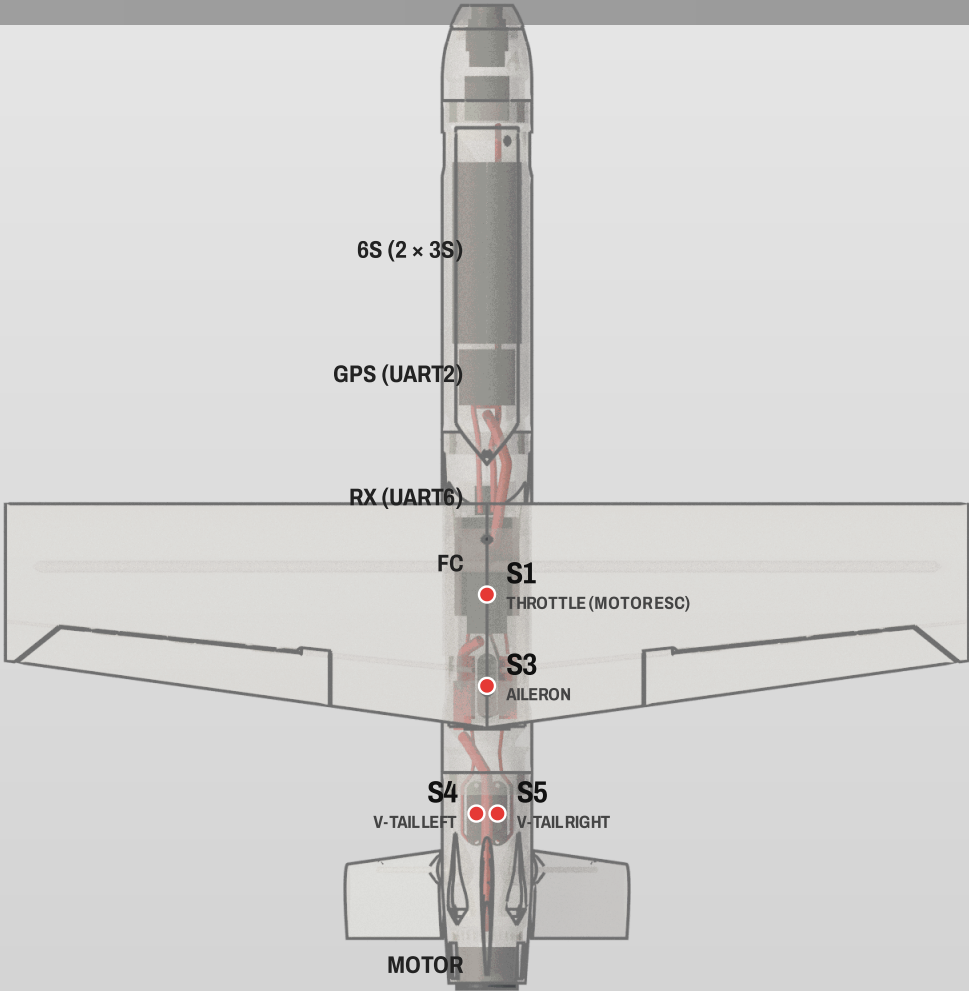


# WIRING DIAGRAM



The diagram shows the flight-controller outputs and where each device sits. Signal, power and video leads follow the routes on the wiring page of the assembly guide; the package has this diagram as a separate large page.

| PAD | FUNCTION             | DEVICE                                          |
|-----|----------------------|-------------------------------------------------|
| S1  | Throttle (motor ESC) | Sequire SQESC 2670 ESC (bidirectional DShot300) |
| S2  | Unused               | Leave empty (DSHOT group with S1)               |
| S3  | Aileron              | Aileron servo (ES9052MD, wing bay)              |
| S4  | V-tail left          | Left tail-panel servo (ES9052MD, tail module)   |
| S5  | V-tail right         | Right tail-panel servo (ES9052MD, tail module)  |



# ARDUPILOT CONFIGURATION

Starting ArduPlane configuration for SkyG Speed 120 on the Matek H743-WLITE (firmware MatekH743-bdshot, ArduPlane 4.5 or later), with the rpm-governor script skyg\_governor.lua. Load the .param file in Mission Planner (Config > Full Parameter List > Load from file), copy the script to the microSD card, reboot, load the .param file again (for the GOV\_ parameters), then work through the bench checks.

Flight controller: on the wing-bay floor on a 0.5 mm foam pad, arrow forward; servo rail Vx set to 6 V for the ES9052MD servos.

**Firmware:** ArduPlane 4.5 or later · **Board:** Matek H743-WLITE (ArduPilot firmware: MatekH743-bdshot)

The setup document and the starting .param file are in the package (ArduPilot folder). Verify every value on the bench before flight.

SKYG

## ARDUPILOT CONFIG

SKYG SPEED 120

MATEK H743-WLITE

# FLIGHT TEST AND SPEED STAGES



The speed envelope opens in stages. Fly each stage only after the ground checks and the previous stage pass: thrust-stand run, wing load and twist test, control freeplay  $\leq 0.2$  mm ( $\leq 0.05$  mm above 45 m/s), aileron hinge stiffness, and the tap test. At each run give short aileron and pitch doublets; any buzz or vibration: throttle to zero, gentle pull-up, land and inspect.

Full procedure, limits and log sheet: the test plan in the Engineering folder.

| Stage Vne | GOV_RPM (full stick) | Expected level speed |
|-----------|----------------------|----------------------|
| 30 m/s    | 10,400 rpm           | 30 m/s               |
| 35 m/s    | 12,000 rpm           | 35 m/s               |
| 40 m/s    | 13,500 rpm           | 40 m/s               |
| 45 m/s    | 15,000 rpm           | 44 m/s               |
| 50 m/s    | 16,600 rpm           | 50 m/s               |
| 55 m/s    | 18,100 rpm           | 54 m/s               |
| 60 m/s    | 19,700 rpm           | 60 m/s               |
| 65 m/s    | 21,200 rpm           | 64 m/s               |
| 70 m/s    | 22,700 rpm           | 69 m/s               |
| 75 m/s    | 24,300 rpm           | 74 m/s               |
| 80 m/s    | 25,800 rpm           | 79 m/s               |
| 85 m/s    | 27,300 rpm           | 84 m/s               |
| 90 m/s    | 28,800 rpm           | 89 m/s               |
| 95 m/s    | 30,300 rpm           | 94 m/s               |
| 100 m/s   | 31,900 rpm           | 99 m/s               |
| 105 m/s   | 33,400 rpm           | 104 m/s              |
| 110 m/s   | 34,900 rpm           | 109 m/s              |
| 115 m/s   | 36,400 rpm           | 114 m/s              |
| 120 m/s   | 37,900 rpm           | 118 m/s              |

**SKYG**

**SKYG SPEED 120**

